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# The consequences of expanded audit reporting: implications of tax key audit matters for tax attribute valuation and auditor-provided tax services

**Daniel P. Lynch**[1] · **Aaron Mandell**[2] · **Linette M. Rousseau**[3]

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## Abstract

This study examines the consequences of tax key audit matters (KAMs) to the valuation of tax attributes and to auditor-provided tax services (APTS). The literature finds that KAM adoption is not associated with investor or audit outcomes, creating an impetus to identify whether specific KAM topics are more consequential. Tax KAMs are an ideal setting because they are prevalent, taxes are economically important, and auditors can provide tax services to their clients. We find that investors discount the tax avoidance and deferred tax assets of tax-KAM companies, suggesting an incentive for managers to avoid tax KAMs. Consistent with self-interest threats to auditor independence, we find that receiving a tax KAM is negatively associated with clients' concurrent purchases of APTS, but that clients increase their future purchases of these services when they stop receiving tax KAMs. These results reveal novel consequences of tax KAMs to the valuation of companies' tax attributes and to APTS incentives.

**Keywords** Tax · Key audit matters · Critical audit matters · KAM · CAM · Auditor-provided tax services

✉ Daniel P. Lynch
daniel.lynch@wisc.edu

Aaron Mandell
amandell@uwm.edu

Linette M. Rousseau
lmrousse2@bauer.uh.edu

1 Wisconsin School of Business, University of Wisconsin-Madison, 975 University Ave, Madison, WI, USA

2 Lubar College of Business, University of Wisconsin-Milwaukee, 3202 N Maryland Ave, Milwaukee, WI, USA

3 Bauer College of Business, University of Houston, 4250 Martin Luther King Blvd, Houston, TX, USA

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## 1 Introduction

This study investigates the consequences of auditors' disclosure of tax key audit matters (KAMs). KAMs are audit report disclosures in which the auditor describes the most challenging audit issues and how the auditor addressed them. KAMs relate to significant risks, transactions, events, estimates, or a combination of these and are intended to provide engagement-specific information to enhance the usefulness of audit reports over the traditional pass/fail audit opinion (IAASB ISA 701).[1] Critics of expanded audit reporting argue that KAMs do not achieve this goal (Minutti-Meza 2021), and many studies show that the adoption of KAM reporting regimes does not affect investors or audit quality (Bédard, Gonthier-Besacier, and Schatt 2019; Gutierrez et al. 2018; Liao, Minutti-Meza, Zhang, and Zou 2022). Research calls for investigation into which KAMs may be consequential (Gutierrez et al. 2018; Minutti-Mezza 2021; Porumb et al. 2021). We answer this call by focusing on the relation between tax KAMs and the specific tax topics they reference and consequences to the market valuation of tax-specific company attributes. We then investigate whether the incentives created by these consequences affect issuers' demand for an important non-audit service, auditor-provided tax services (APTS). This allows us to examine the effect of KAM reporting in a manner tightly mapped to the content of the KAMs.

Tax KAMs provide an excellent setting to examine whether KAM disclosures have consequences to the client company and to the purchase of non-audit services for several reasons. Tax KAMs are prevalent, tax expense is material to most companies' financial statements, there is substantial risk and subjectivity associated with the tax function (Deloitte 2011; Plumlee and Yohn 2010), and tax KAMs discuss a diverse range of issues. Taxes are among the most common types of KAMs, with 43% of companies in the FTSE 350 receiving a tax KAM (FRC 2016b). There is also significant variation in the specific issues in tax KAMs, with uncertain tax positions, deferred tax assets, and tax complexity being the most common issues referenced. Research on KAM types does not consider this variation. We fill this void by examining whether specific issues referenced in tax KAMs have consequences for those specific tax attributes.

Additionally, taxes are unique because auditors can provide tax services to their clients, whereas they cannot provide accounting services to clients related to other areas of financial reporting. This creates a potential unintended consequence of tax KAMs related to incentives around APTS. Given the recent resurgence in non-audit services (Beardsley, Lassila, Omer 2019; Michaels 2022), it is important to

[1] The United Kingdom implemented expanded audit reporting for companies with premium listings on the London Stock Exchange in 2013 and adopted KAM reporting for all companies in 2017. The PCAOB adopted similar critical audit matter (CAM) standards in 2019. For clarity and because we use UK data, we use the acronym KAMs throughout this paper.

understand whether fees for non-audit services create perverse incentives in KAM reporting decisions. The unique incentives around tax KAMs, combined with our ability to observe tax complexity, the valuation of tax avoidance and deferred tax assets, and APTS engagements, allows us to investigate the effect of KAM reporting in a manner tightly mapped to the content of the KAMs.

Our sample comprises 3,036 company-year observations from companies traded on the London Stock Exchange Main Market premium segment from 2013–2019. We use the UK setting because the country's early adoption of KAM reporting provides the time-series data required to investigate the consequences of receiving and resolving tax KAMs, in contrast to the one and a half years of data available between the June 2019 adoption of critical audit matter (CAM) reporting in the United States and the onset of the COVID- 19 pandemic, which could confound results in the US setting. Further, selecting an English-speaking setting allows us to manually review the tax KAMs and hand-code them based on the specific tax issues they address.

We first examine whether investors differentially value the tax attributes of tax KAM companies. The literature suggests investors positively value both tax avoidance (as it increases after-tax cash flows) and deferred tax assets (as they represent future tax deductions to the company) (Desai and Dharmapala 2009; Wilson 2009; Laux 2013; Inger 2014). However, because of the uncertainty related to tax avoidance and the realizability of deferred tax assets, along with the lag between recording these benefits and their final realization, these positive valuations are attenuated under certain conditions (Crocker and Slemrod 2005; Desai and Dharmapala 2009; Balakrishnan et al. 2019; Inger 2014; Jacob and Schutt 2020; Guenther and Sansing 2000; Laux 2013; Flagmeier 2022). We expect that tax KAMs provide a confirmatory signal about the riskiness of the company's tax positions and deferred tax assets.

Consistent with this conjecture, we find that receiving a tax KAM attenuates the positive market valuation of companies' tax avoidance and deferred tax assets. We also confirm that the deferred tax assets of non-tax KAM companies with a high predicted value of receiving a tax KAM (based on an expectations model) are not attenuated like those of tax KAM companies.[2] These findings are robust to controlling for the quality of tax footnote disclosures (Luo et al. 2023). Collectively, these results suggest that tax KAMs signal the riskiness of companies' tax avoidance activities, the realizability of deferred tax assets, or both, leading to a lower valuation of these tax attributes. Importantly, this signal does not need to be due to new information. Rather, a tax KAM can confirm management's prior tax disclosures from the auditor, who has greater source credibility than management (Christensen et al. 2014; Gimbar et al. 2016). Collectively, the negative valuation consequences of receiving a tax KAM suggest that management has incentives to avoid tax KAM disclosure where possible. Illustrating the sensitivity of tax KAMs, one audit partner states: "My biggest fear is that people are making inaccurate assumptions that

[2] We develop an expectations model to predict the likelihood a company will receive a tax KAM using a probit regression of tax KAM issuance on an array of company economic characteristics. See Appendix A for detailed methodology and results.

causes some sort of reputational damage to the company … that someone could read into that [tax KAM] and say … 'They're not a good tax paying citizen'" (Griffith, Rousseau, and Zehms 2024).

We next examine whether these incentives affect client companies' purchases of auditor-provided tax services. There are two reasons to anticipate that companies can avoid receiving tax KAMs by purchasing these services. First, the revenue stream from APTS forms an economic bond between the auditor and the client that creates an incentive for the auditor to acquiesce to the client's preferred reporting (Coffee 2006; Alsadoun et al. 2018; Causholli et al. 2014; Beardsley et al. 2019, 2021)—in this setting, deciding not to issue a tax KAM, even when it is warranted. Second, the APTS engagement could enhance the auditor's expertise about the nuances of the client's tax function, generating knowledge spillovers that make taxes less challenging to audit and less likely to be disclosed as a KAM. Accordingly, we predict and find that receiving a tax KAM is negatively associated with clients' concurrent APTS purchases. Specifically, companies that continue to receive tax KAMs decrease their APTS fees by 32 percent (about \$63,000 US dollars), while clients with unexpectedly low levels of tax KAM disclosure (based on our expectations model) purchase more APTS. This suggests that higher fees for these services allow clients to avoid receiving tax KAMs, either due to economic bond threats to independence or to quality-increasing knowledge spillovers. Given that KAM reporting is not associated with changes in audit fees (Bédard et al. 2019; Gutierrez et al. 2018), these results illustrate a topic-specific effect of tax KAMs on auditor compensation.

To identify which competing explanation drives our results, we examine the timing of changes in APTS in companies that resolve (i.e., stop receiving) tax KAMs. A positive *contemporaneous* association between purchases of APTS and tax KAM resolution is consistent with either knowledge spillovers or self-review bias (Sun and Habib 2021).[3] In contrast, a positive association between resolving tax KAMs in the *current* period and increasing APTS in the *future* is consistent with economic bond incentives that may impair auditor independence (DeAngelo 1981). As Coffee (2006, p. 66) states: "The real conflict lies not in the actual receipt of high fees, but in their expected receipt. Even the client currently paying low consulting revenues to its auditor might reverse this pattern if the auditor proved more cooperative." Accordingly, we find that companies that resolve tax KAMs do not change APTS in the year the tax KAM is resolved but increase their purchases of APTS by 104 percent (about \$208,000 US dollars) the following year. This suggests that auditors that stop disclosing tax KAMs are rewarded with higher APTS fees.[4] These findings are consistent with economic bond threats to independence.

[3] Self-review bias refers to the threat to independence that arises when an auditor audits their own work. For example, an auditor may lack independence when auditing the financial accounting treatment for an uncertain tax position that the auditor recommended to the client (Sun and Habib 2021).

[4] We do not find any support for the alternative explanation that companies change their tax avoidance in response to tax KAMs. This suggests either managers do not believe the costs of tax KAM disclosure outweigh the benefits of tax avoidance or reducing tax avoidance, tax complexity, or both (e.g. multinational corporations subject to many tax jurisdictions) (Balakrishnan et al. 2019; Kim et al. 2019). See Sect. 5.4 for a description of this analysis.

To enhance our contribution regarding settings in which KAMs are consequential, we separately examine the specific tax issues cited in each tax KAM. After noting inaccuracies in Audit Analytics' classifications of tax KAM topics (see Appendix D), we read each tax KAM in our sample and categorize each one into four primary and nonmutually exclusive reasons: (1) uncertain tax positions (78%), (2) deferred taxes (33%), (3) multijurisdictional complexity (60%), and (4) other (5%). We disaggregate the tax KAMs in our sample into these four subtypes and test how closely companies' tax fundamentals map into auditors' disclosures. We find that companies' tax avoidance (lower effective tax rate) is the strongest predictor of receiving an uncertain tax position KAM. We find that higher effective tax rate volatility and larger deferred tax asset balances are positively associated with receiving KAMs related to deferred taxes. Tax KAMs related to multijurisdictional complexity are more common in companies with larger foreign operations. Together, these results illustrate that tax KAMs are associated with specific information about companies' tax activities.

We then repeat our consequences analysis for each specific tax KAM topic. We find that receiving an uncertain tax position KAM is associated with lower valuations of both tax avoidance and deferred tax assets, while deferred tax asset KAMs are only associated with lower deferred tax asset valuation. Further, companies receiving an uncertain tax position KAM have lower valued tax avoidance than do companies receiving a deferred tax asset KAM. This suggests that the specific discussion in each tax KAM is associated with varied valuation consequences. Concerning consequences to non-audit services, we find that the association between resolving a tax KAM and increased APTS is strongest for deferred tax asset and uncertain tax position KAMs, where the auditor has the most discretion over whether to issue the KAM.[5] Collectively, these results reinforce our primary inferences concerning economic bond threats to independence.

Our study makes several contributions to research and practice. First, we demonstrate that tax KAMs and the specific tax issues the auditor describes within them are associated with how investors value companies' tax attributes. This result highlights the importance of examining constructs related to the topical content of KAMs in researching expanded audit reporting. Furthermore, we show that KAM consequences can vary with the specific issues cited within a single type of KAM. This highlights the need to examine specific KAM content and not rely on subtypes categorized by data providers to understand the consequences of expanded audit reporting. Second, we reveal that a consequence of KAM reporting is to increase companies' demand for APTS. The potential of non-audit service fees to impair auditor independence is an area of renewed concern in the literature and for regulators and other stakeholders in the United Kingdom and the United States

[5] We confirmed with practitioners that certain types of tax KAMs are difficult to resolve (i.e., complexity due to the size of foreign operations), while those related to specific uncertain tax positions and the realizability of deferred tax assets are more likely to change and be resolved.

(Financial Reporting Council 2019; Rapoport 2018; Michaels 2022; Carcello et al. 2020; Beardsley et al. 2019, 2021; Blay and Geiger 2013; Donelson et al. 2020). We extend this literature by illustrating the implications of APTS for auditor independence in KAM disclosure decisions.

Third, our results will interest policymakers as they evaluate the effectiveness of expanded audit reporting requirements. Despite the intent of KAM standards, there is widespread doubt that KAMs contain decision-useful information (Lennox, Schmidt, Thompson 2023; Minutti-Meza 2021) and a body of archival evidence indicating that the adoption of KAM reporting regimes did not affect audit outcomes or investor decision-making (Bédard et al. 2019; Gutierrez et al. 2018; Liao et al. 2022). However, much of this literature treats all KAMs the same without exploring differential effects across their topics. We answer the call of Minutti-Mezza (2021, p. 42) for further research "determining whether common KAMs topics have unique effects on complex aspects of financial reporting" and "developing an expectations model for KAMs."[6] Our study emphasizes the importance of focusing on settings relevant to the specific KAM topic in making valid inferences about KAMs.

## 2 Background and hypothesis development

### 2.1 Key Audit Matter institutional setting

KAMs are audit report disclosures where the auditor describes the most significant issues encountered in the audit and how the auditor addressed them (IAASB ISA 701–8). KAMs typically relate to significant risks, transactions, events, estimates, or a combination of these (IAASB ISA 701–9).[7] James Gunn, managing director of Professional Standards for the IAASB, summarizes the purpose of KAM reporting as to "enable auditors to be as entity-specific and audit-specific as possible in describing each KAM so that they continue to be relevant and useful to investors and other users," as opposed to the former audit reporting model in which the auditor used standardized language to convey a pass/fail opinion. KAMs do not change or disclaim the auditor's overall pass/fail opinion on the financial statements. The Financial Reporting Council regulates auditors in the United Kingdom and began requiring auditors to report KAM-like risk of material misstatement disclosures for London Stock Exchange premium listed companies in 2013. The Financial Reporting Council revised those disclosures for greater conformity with IAASB KAM standards and expanded them to include all public companies in 2017. Almost all

[6] Gutierrez et al. (2018) echo this call for research on specific KAM topics. Burke et al.'s (2023) finding that unexpected revenue KAM disclosures influence the market further motivates this line of inquiry.

[7] See Appendix B for an example of a KAM from KPMG's 2019 audit report for Unilever PLC.

companies trading on the London Stock Exchange receive at least one KAM, and the average audit report contains three KAMs.[8]

A significant literature addresses the effects of KAMs on various investor and audit outcomes. These papers investigate whether adopting KAM reporting affects investor decision-making, financial reporting quality, audit quality, audit fees, and audit report delay. The results are mixed. Bédard et al. (2019), Gutierrez et al. (2018), Liao et al. (2022), and Burke et al. (2023) find that KAM reporting regimes in France, the United Kingdom, China/Hong Kong, and the United States, respectively, do not affect investors' market activity, audit quality, or audit pricing. Conversely, other studies using UK data find that KAM reporting prompts stock market reactions over long windows (Lennox et al. 2023), affects loan contracting (Porumb et al. 2021), and improves financial reporting quality without affecting audit fees or report delay (Reid, Carcello, Li, and Neal 2019).

Limited concurrent research examines KAMs by topical area. Rousseau and Zehms (2024) and Rousseau (2024) document that UK audit partners and audit committees, respectively, influence KAM reporting topical diversity. Andreicovici et al. (2023) find that companies receiving goodwill-related KAMs expand management's disclosures around goodwill risks and make more timely goodwill impairments the next year. Abbott and Buslepp (2023) and Nylen, Wangerin, and Zehms (2024) show that merger and acquisition-related KAMs inform audit report users. Burke et al. (2023) examine the predictors of various topics of US CAMs and generally find that the materiality of CAM-related financial statement line items and the receipt of CAM topic-related SEC comment letters are the most important CAM determinants. Finally, concurrent research shows that tax KAMs are negatively associated with tax-related earnings management (Drake et al. 2024) and that auditors are less likely to issue tax KAMs when perceived tax enforcement is high (Filosa et al. 2024).

### 2.2 What makes tax KAMs special?

Despite the literature demonstrating that, on average, KAMs do not affect market and auditor outcomes (Bédard et al. 2019; Gutierrez et al. 2018; Burke et al. 2023), tax KAMs disclosure could be of greater than average concern to auditors and clients. Tax KAMs can expose the auditor's challenges in gaining comfort over the client's tax accounting and interactions with tax authorities, emphasizing uncertainty about the realizability of the client's uncertain tax positions and deferred tax assets to regulators and financial statement users. Further, tax KAMs may highlight clients' aggressive tax avoidance strategies, risking negative political and reputational scrutiny around whether the client pays its fair share of taxes (Austin and Wilson 2017; Barford and Holt 2013; Fox and Wilson 2023; Graham et al. 2014; Griffith et al. 2024; Scholes et al. 2014; Wilde and Wilson 2018). Tax KAMs also reduce managers' discretion to manage earnings through the tax account (Drake et al. 2024),

[8] PCAOB Critical Audit Matter (CAM) disclosure requirements became effective in June of 2019 and are very similar to FRC KAMs. The major difference between the two regimes is that PCAOB CAMs must "relate to accounts or disclosures that are material to the financial statements" (PCAOB AS 3101–11). As a result, the average number of CAMs for US issuers is 1.67 (Coleman, Conley, and Hallas 2021).

closing off an avenue for last-chance earnings management (Dhaliwal et al. 2004). Finally, tax KAMs are unique in that auditors can provide tax compliance and consulting services to their audit clients, whereas they cannot provide most services for other accounts. This creates a potential unintended consequence of KAM reporting to incentives to purchase APTS. Collectively, these factors suggest that clients could be more concerned with tax KAMs than other KAM topics, with greater incentives to avoid receiving a tax KAM if possible.

### 2.3 Potential consequences of tax KAMs to the valuation of client tax attributes

The consequences of tax KAMs to the valuation of companies' tax attributes depends on how investors perceive tax KAMs. As KAMs are not intended to communicate original information about the company (ISA 701), investors may not change how they value its tax attributes. However, auditors' secondary discussions of management's tax disclosures could provide a confirmatory risk signal to investors, who view auditors as highly credible (Christensen et al. 2014; Gimbar et al. 2016). To the extent tax KAMs signal investors about the riskiness of certain tax attributes or increase the salience of other tax disclosures in the financial statements, we could observe valuation effects. We therefore examine the valuation of two important tax attributes of the company: (1) tax avoidance and (2) deferred tax assets.

Typically, shareholders positively value corporate tax avoidance because it increases after-tax cash flows (Wilson 2009; Hanlon and Heitzman 2010; Desai and Dharamapala 2009). However, when tax avoidance reduces corporate transparency and creates agency costs, this positive valuation is attenuated (Crocker and Slemrod 2005; Desai and Dharamapala 2009; Hanlon and Slemrod 2009; Balakrishnan et al. 2019; Hope et al. 2013; Luo et al. 2023). The level of risk associated with tax avoidance varies across different types of tax planning measures. The tax avoidance continuum ranges from tax evasion, such as tax shelters, to benign avoidance, such as investing in tax-exempt municipal bonds (Hanlon and Heitzman 2010; Inger 2014). However, shareholders typically have insufficient information to determine the riskiness of a company's tax planning (Crocker and Slemrod 2005; Balakrishnan et al. 2019). There is also a significant lag between when a company takes a tax position on a filed tax return and the ultimate audit and settlement of the tax position with the tax authority, which creates additional uncertainty. Finally, investors discount higher-risk tax avoidance strategies when information asymmetry is high, consistent with greater uncertainty about the realizability of the tax benefits (Wilson 2009; Hanlon and Slemrod 2009; Inger 2014; Drake et al. 2019). To the extent tax KAMs signal the riskiness of underlying tax avoidance strategies, we expect to observe a lower valuation of tax KAM companies' tax avoidance. We therefore predict:

*H1a: Tax KAMs are negatively associated with the valuation of tax avoidance.*

Tax KAMs may also affect the valuation of deferred tax assets. Deferred tax assets represent future tax deductions that are only realized if the company generates future taxable income. Common items that create deferred tax assets are loss carryforwards, accruals of reserves, and accruals of post-employment benefits.[9] Under IFRS, deferred tax assets are recorded only to the extent it is "probable" that future taxable income will allow the deferred tax asset to be used (IAS 12). Determining whether the company will generate future taxable income is a subjective assessment that involves significant manager judgment, making deferred tax assets difficult to audit. Research suggests that, due to this inherent subjectivity and the lag between when deferred tax assets are generated and realized, there is heterogeneity in how these assets affect future cash taxes and how investors value them (Flagmeier 2022; Amir et al. 1997; Guenther and Sansing 2000; Laux 2013; Amir and Sougiannis 1999).

Tax KAMs could reflect the uncertainty in the realizability of deferred tax assets by revealing the nature and extent of complex estimates the auditor faced in assuring the deferred tax asset balance. For example, many tax KAMs use language to the effect that estimation uncertainty means that tax balances have "a potential range of reasonable outcomes greater than our materiality for the financial statements as a whole" (KPMG 2018, p. 69). If the auditor's decision to issue tax KAMs confirms the uncertainty about the realizability of deferred tax assets, we expect that the market will discount the deferred tax assets of tax KAM companies. We therefore predict:

*H1b: Tax KAMs are negatively associated with the valuation of deferred tax assets.*

Despite tax KAMs' potential to highlight uncertainty about companies' future tax benefits, tax KAMs may not influence the valuation of companies' tax attributes. If tax KAMs do not provide any signal to investors regarding the company's tax avoidance strategies or the realizability of its deferred tax assets, we would not expect valuation differences. This will be the case if the inputs to the auditors' KAM decisions are divorced from the information the market finds relevant when evaluating companies' tax accounts (i.e., based on audit effort and hours rather than estimation uncertainty). Further, Lennox et al. (2023) suggest that KAMs do not move markets because investors fully incorporate the information contained in KAMs before they are disclosed.

### 2.4 Potential consequences of tax KAMs to auditor-provided tax services

The consequences of tax KAMs to APTS will depend on management's perception of how audit report users will respond to tax KAMs. To the extent tax KAMs are associated with the valuation of specific tax attributes, management could have incentives to avoid tax KAM disclosure. Ways to avoid tax KAMs could include purchasing APTS to avoid initial disclosure of tax KAMs, investing in these services to resolve existing tax KAMs, economically punishing auditors that issue tax

[9] Under U.S. GAAP, firms record a valuation allowance for deferred tax assets that are "more likely than not" to not be realized. Firms do not report a valuation allowance under IFRS.

KAMs, or a combination of these. We therefore investigate the consequences of tax KAMs to clients' purchases of APTS.

The UK Financial Reporting Council's Ethical Standard (2016a) permits a variety of APTS, including advice on specific issues, tax planning, compliance, and organizational structures, subject to limitations and safeguards designed to preserve auditor independence (Financial Reporting Council 2016c). These safeguards can include having non-engagement team personnel conduct the tax work, independent review by a senior tax employee or partner, obtaining external advice on the work, or a combination of these (FRC 2016c). The audit firm has discretion to determine whether to employ safeguards and which ones on a case-by-case basis (Financial Reporting Council 2016c). Collectively, these regulations afford the audit firm substantial discretion in deciding which APTS to offer and which, if any, independence safeguards are necessary. While many studies conducted following the Sarbanes–Oxley Act of 2002 found that non-audit services did not affect auditor independence (Ashbaugh et al. 2003; Chung and Kallapur 2003; Defond et al. 2002; Reynolds, Deis, and Francis 2004), more recent research suggests that the revival of large advisory practices in audit firms compromises audit independence and quality (Carcello et al. 2020; Beardsley et al. 2019, 2021; Blay and Geiger 2013; Donelson et al. 2020).

The literature suggests two potential ways client companies could avoid tax KAMs through APTS. First, the services could create knowledge spillovers that reduce the complexity of auditing the tax account "by accelerating audit firm awareness of transactions material to the financial statements" (De Simone et al. 2015, p. 1472). Consistent with APTS creating quality-increasing audit efficiencies, research shows that APTS are negatively associated with internal control material weaknesses and are positively associated with financial reporting quality (De Simone et al. 2015; Kinney, Palmrose, and Scholz 2004; Paterson and Valencia 2011; Gleason and Mills 2011), accurate going concern opinion reporting (Robinson 2008), and with the use of more effective audit procedures (Joe and Vandervelde 2007).

Conversely, purchases of APTS could help managers avoid tax KAMs by compromising the auditor's independence via self-interest and self-review bias (Sun and Habib 2021). APTS are unique because the auditor can provide tax services and subsequently audit the tax provision. This contrasts with other areas of financial reporting (and KAM types), where standards forbid the auditor from providing both assurance and consulting services. The fees from APTS form an economic bond between the auditor and the client, creating perverse incentives for the auditor to acquiesce to the client's preferred reporting (Coffee 2006). Accordingly, the literature finds that fees for non-audit service are associated with lower actual and perceived audit quality (Alsadoun et al. 2018; Causholli et al. 2014; Srinidhi and Gul 2007; Beardsley et al. 2019, 2021). Additionally, the auditor may be less willing to issue a KAM stating that tax activities their firm advised on are especially risky, complex, and subjective. Consistent with this view, Choudhary, Koester, and Pawlewicz (2022) find that clients with material purchases of APTS have lower tax accrual quality and that this effect is concentrated among audit offices where self-review threats are strongest. In sum, clients may be able to avoid tax KAMs with APTS, either through knowledge spillover, economic bond bias, or self-review bias.

It follows that, if clients economically reward auditors that do not issue tax KAMs by purchasing more with higher APTS, they may similarly punish auditors by reducing their purchases if a tax KAM is disclosed. Specifically, the client could experience frustration that the auditor considers the tax account especially risky, despite the audit firm's involvement with the tax function. Management may also be concerned that the additional insight and information the auditor gains through the provision of APTS contributes to the decision to issue a tax KAM. Consistent with this view, Cade and Hodge (2014) and Gay and Ng (2015) provide experimental evidence that management and the audit committee, respectively, communicate less openly with the auditor under KAM reporting regimes. Similar reward and punishment responses may exist on the part of the auditor. Specifically, an auditor upset at a client's decision to reduce APTS may respond by issuing more tax KAMs. We therefore predict:

> *H2a: Receiving a tax KAM is negatively associated with companies' concurrent purchases of auditor-provided tax services.*

We next distinguish between the knowledge spillover, self-review threat, and self-interest threat explanations for the association between tax KAMs and APTS by examining clients that resolve (i.e., stop receiving) tax KAMs. The timing of tax KAM resolution relative to the timing of changes in APTS provides insight into which mechanism is at work. A positive association between the *contemporaneous* purchases of APTS and tax KAM resolution is consistent with either knowledge spillovers or self-review bias, as the fees reflect tax work completed during the period of KAM resolution. In contrast, a positive association between resolving tax KAMs in the *current* period and increasing APTS in the *future* is consistent with impaired independence due to economic bonding.[10] This delay is expected for self-interest threats because the future *multi*-period revenues the auditor can ensure by retaining the client are more significant to the auditor's overall economic interest than the current *single*-period fee (DeAngelo 1981). Accordingly, "the real conflict lies not in the actual receipt of high fees, but in their

[10] We spoke with practitioners about the timing of negotiating engagements to provide tax services relative to the timing of the financial statement audit (and KAM decision). The practitioners shared that new APTS engagements typically begin in the year following the successful sale of APTS. For example, one practitioner stated that, during a 2022 calendar year-end audit of a client that took place during February of 2023, the audit firm successfully acquired the tax work of the client for 2023. This practitioner stated the firm would not take over the 2022 tax work as the existing tax service provider had already engaged in tax planning and compliance activities for the year. Given we measure both tax KAMs and APTS based on the annual financial statements, this implies that the auditor knows about a new APTS engagement at time t, but does not perform that engagement and receive associated fees until time t + 1. Collectively, this evidence helps us attribute subsequent purchases of APTS following tax KAM resolution to economic bond concerns rather than knowledge spillovers.

expected receipt. Even the client currently paying low consulting revenues to its auditor might reverse this pattern if the auditor proved more cooperative" (Coffee 2006, p. 66). Consistent with the potential of future fees to compromise auditor independence, Causholli et al. (2014) find that auditors permit clients with high growth in non-audit service fees from *t* to *t +1* to manage earnings more aggressively at time *t,* before any money has changed hands. Similarly, Blay and Geiger (2013) find that auditors issue fewer going concern opinions for clients that pay higher *future* fees. We therefore predict:

*H2b: Resolving a tax KAM at time t is positively associated with companies' future purchases of auditor-provided tax services at time t + 1.*

There are several reasons we may fail to find support for H2. First, because a KAM does not modify or disclaim a clean audit opinion, management may be indifferent to receiving a tax KAM and see no reason to alter its purchases of APTS. Second, management could reduce the company's tax complexity such that the auditor no longer considers the tax provision among the most complex audit issues (e.g., reduce tax avoidance or not realize a deferred tax asset on the balance sheet). Third, the complexity that gave rise to the tax KAM could resolve organically without the auditor or management taking action. For example, the statute of limitations could expire on an uncertain tax position that led to a tax KAM or a company may generate taxable income and therefore recognize a DTA.

## 3 Research design

### 3.1 Sample selection

Our starting sample includes London Stock Exchange premium segment companies with at least one KAM during the period 2013 to 2019 in Audit Analytics.[11,12,13] We exclude companies that lack FactSet International data required

[11] This number includes risk of material misstatement disclosures, which the Financial Reporting Council required for premium listed companies beginning in 2013. These disclosures and KAM requirements are similar, and considering them together is the standard in the literature examining expanded audit reporting in the United Kingdom (Rousseau and Zehms 2024; Gutierrez et al. 2018; Reid et al. 2019).

[12] Only 27 audit opinions for London Stock Exchange firms during our sample period report zero KAMs.

[13] While the US Tax Cuts and Jobs Act (TCJA) was introduced during our sample period, it did not significantly affect the frequency of tax KAMs (See Table 7 Panel A). A review of the text of our tax KAMs shows that only nine of them specifically mention the TCJA. Eight of these also describe unrelated complexities arising from transfer pricing and uncertain tax positions in non-US jurisdictions, indicating that the TCJA was not the sole basis for issuing the tax KAM.

**Table 1** Sample Selection

| | | |
|---|---|---|
| **Panel A: Valuation Testing** | | |
| Total company-year observations | | 7,339 |
| Less: | Observations without Factset identifiers | (1,012) |
| | Missing currency conversion data | (61) |
| | Missing Book ETR data | (11) |
| | Missing financial statement controls | (373) |
| | Missing lagged data for multi-year variables | (2,721) |
| | Non-Main Market, premium segment firms | (98) |
| Sample for valuation model | | 3,063 |
| Number of unique companies included in valuation model | | 601 |
| **Panel B: Auditor-Provided Tax Services Testing** | | |
| Sample for valuation model | | 3,063 |
| Less: | Observations without lagged data required to calculate change variables | (217) |
| Sample for tests of changes in tax KAM disclosure and auditor-provided tax services | | 2,846 |
| Number of unique companies included in auditor-provided tax services testing | | 699 |

This table reports the construction of our samples for the valuation of tax attributes (Model (1)) and for auditor-provided tax services testing (Models (2)–(4))

to compute effective tax rates or our financial statement controls. This results in a final sample of 3,063 company-year observations, including 564 observations with at least one tax key audit matter. Table 1 describes our sample construction.[14] Table 2 describes the distribution of tax KAMs over time and by industry. Tax KAMs and each subtype of tax KAM are roughly evenly distributed across time, except for 2013—the first, partial year of expanded disclosure. We discuss our categorization of tax KAM subtypes compared to Audit Analytics' classification and the distribution of KAM subtypes within our sample in Sect. 5.1.

[14] The UK tax system is relatively stable during our sample period. The corporate tax rate is 20% for years 2014–2016 and 19% from 2017 onward. The UK implemented a territorial tax system in 2009. The only tax-related accounting change during our sample period is the IASB's 2019 adoption of IFRIC 23, "Uncertainty over income tax treatments." Our results are robust to excluding observations from 2019.

**Table 2** Frequency of Tax KAMs

**Panel A: Frequency of Tax KAMs by Year**

| | **Number of Tax KAMs** | | | | |
|---|---|---|---|---|---|
| **Year** | **Total** | **UTP** | **DTA** | **Complexity** | **Other** |
| 2013 | 58 | 40 | 19 | 23 | 3 |
| 2014 | 102 | 73 | 36 | 47 | 3 |
| 2015 | 84 | 70 | 29 | 53 | 1 |
| 2016 | 88 | 72 | 32 | 58 | 2 |
| 2017 | 89 | 67 | 30 | 59 | 5 |
| 2018 | 79 | 64 | 26 | 55 | 4 |
| 2019 | 64 | 56 | 14 | 45 | 10 |
| Total | 564 | 442 | 186 | 340 | 28 |

**Panel B: Frequency of Tax KAMs by Industry**

| | **Number of Tax KAMs** | | | | |
|---|---|---|---|---|---|
| **Industry classification** | **Total** | **UTP** | **DTA** | **Complexity** | **Other** |
| Consumer Nondurables | 72 | 63 | 19 | 52 | 5 |
| Consumer Durables | 3 | 3 | 3 | 3 | 0 |
| Manufacturing | 62 | 52 | 25 | 43 | 3 |
| Oil, Gas, and Coal | 16 | 10 | 6 | 8 | 0 |
| Chemicals | 34 | 26 | 3 | 29 | 0 |
| Business Equipment | 43 | 33 | 16 | 31 | 0 |
| Telephone and TV | 12 | 12 | 11 | 7 | 0 |
| Utilities | 4 | 4 | 0 | 0 | 0 |
| Wholesale & Retail | 58 | 45 | 13 | 37 | 0 |
| Health, Medical, and Drug | 27 | 25 | 8 | 26 | 2 |
| Finance | 74 | 45 | 22 | 10 | 13 |
| Other | 159 | 124 | 60 | 94 | 5 |
| Total | 564 | 442 | 186 | 340 | 28 |

This table presents counts of tax KAMs from our sample of 3,063 company-year observations. Panel A reports the distribution of KAMs, in total and by type by year. Panel B reports this count by industry, using Fama and French 12 industry classifications

## 3.2 Testing the consequences of tax KAMs to valuation of companies' tax attributes

To examine whether receiving a tax KAM attenuates the positive valuation of companies' tax attributes, we first compare valuations for tax KAM and non-tax KAM companies and estimate the following fully interacted ordinary least squares (OLS) regression for each subsample:

$$\begin{aligned} TOBIN'SQ_{i,t+1q} = {} & \gamma_0 + \gamma_1 AVOID_{i,t} + \gamma_2 DTA\_NET_{i,t} + \gamma_3 ETR_{VOL_{i,t}} \\ & + \gamma_4 HAVEN_{i,t} + \gamma_5 FOREIGN_{i,t} + \gamma_6 PRIORLOSS_{i,t} \\ & + \gamma_7 MATERIALITY_{,t} + \gamma_7 NOL_{i,t} + \gamma_8 SALES_{i,t} \\ & + \gamma_9 SALES_{GROWTH_{i,t}} + \gamma_{10} RETURN\_VOL_{i,t} \\ & + \gamma_{11} LEVERAGE_{i,t} + \gamma_{12} INTANGIBLE_{i,t} \\ & + \gamma_{13} FCF_{,t} + \gamma_{14} R\&D_{i,t} + \gamma_{15} CAPEX_{,t} + \gamma_{16} ACCRUALS_{i,t} \\ & + \mu_{ind} + \tau_t + \varepsilon_{i,t}. \end{aligned} \tag{1}$$

Next we use Model (1) to compare tax KAM companies to non-tax KAM companies that have a 75 percent or greater probability of receiving a tax KAM based on their economic fundamentals obtained using the expectations Model (5) in Appendix A.

Following the literature, we use Tobin's Q to proxy for firm value (Desai and Dharmapala 2009; Luo et al. 2023). To ensure that any tax KAM disclosure is imputed into the company's valuation, we measure Tobin's Q at the first quarterly or semi-annual reporting date following the issuance of the company's annual report.[15],[16] To measure tax avoidance, we compute the one-year book effective tax rate as income tax expense divided by pretax book income.[17] To ease interpretation, we multiply this effective tax rate measure by −1 to generate the variable *AVOID*, such that higher values are associated with higher levels of tax avoidance. *DTA_NET* equals total deferred tax assets less total deferred tax liabilities, scaled by total assets.[18] ETR volatility (*ETR_VOL*) is the standard deviation of ETR over the period t - 2 through t. To measure tax complexity related to foreign operations, we include *FOREIGN*, which is equal to the percentage of foreign sales and *HAVEN*, an indicator variable equal to one if the issuer is headquartered in one of the tax haven countries designated by Dyreng and Lindsey (2009).

We include several other potential predictors of tax KAMs. Research suggests that the materiality of the related account primarily determines the CAM topics disclosed in the United States. (Burke et al. 2023). Therefore, we include *MATERIALITY*, equal to income tax expense scaled by company assets.[19] We control for net operating losses with the indicator *NOL*, which is equal to one if the company has negative income tax expense and zero otherwise, and with *PRIORLOSS*, an indicator variable equal to one if the company reported net losses during the

[15] Our findings are consistent when measuring annual Tobin's Q contemporaneously with KAM disclosure, following Luo et al., (2023).
[16] The United Kingdom revoked its quarterly reporting requirement in 2014 following the European Commission's call to adopt measures designed to reduce short-termism in financial reporting (European Commission 2013; Financial Conduct Authority 2014). Accordingly, the first reporting interval after tax KAM issuance for many of our sample firms is semi-annual.

[17] We winsorize effective tax rate measures at zero and one. We set effective tax rates of loss firms with negative (positive) tax expense equal to 0 (1) (Edwards, Schwab, and Shevlin 2016).
[18] We do not include valuation allowances in our analyses because under IFRS deferred tax assets are reported at their probable net realizable value (i.e., there is no valuation allowance account).

[19] We acknowledge that the independent variables *ETR* and *MATERIALITY* are related constructs as they both include tax expense. All expectation model results are robust to omitting *MATERIALITY* (untabulated).

current year or prior five years. We include these predictors of tax KAMs as they likely affect the valuation of tax attributes (Drake et al. 2019) and correct for any remaining covariate imbalance in our matched sample tests (Shipman et al. 2017). Following research on the valuation of tax avoidance (Desai and Dharmapala 2009; Inger 2014; Drake et al. 2019; Luo et al. 2023), we control for company characteristics associated with firm value, including the natural log of sales (*SALES*), sales growth (*SALES_GROWTH*), company risk (*RETURN_VOL*), total leverage (*LEVERAGE*), intangible asset intensity (*INTANGIBLE*), free cash flow (*FCF*), research and development (*R&D*), capital expenditures (*CAPEX*), discretionary accruals (*ACCRUALS*), and trading volume volatility (*VOLUME_VOL*). We present detailed variable definitions in Appendix C. We winsorize all continuous variables at 1 percent and 99 percent. We employ robust standard errors clustered at the company level in all analyses.[20]

### 3.3 Testing the consequences of tax KAMs to auditor-provided tax services

We test H2a, that receiving a tax KAM is negatively associated with clients' *concurrent* purchases of APTS in two ways. First, we examine whether clients receiving tax KAMs reduce these purchases using the following OLS regression:

$$\begin{aligned}\Delta APTS_{i,t-1,t} = \beta_0 + \beta_1 TAXKAM_{FIRM_{i,tt-1}} + \beta_2 RESOLVE\_TAXKAM_{i,t} \\ + \Delta X_{i,t-1,t}/\Delta X_{i,t,t+1} + \mu_{ind} + \tau_t + \varepsilon_{i,t}.\end{aligned} \tag{2}$$

We operationalize the dependent variable, *ΔAPTS*, as the raw change in APTS (*ΔLN_APTS*) and the percentage change in APTS (*%ΔAPTS*). The independent variable of interest is $TAXKAM\_FIRM_{i,t\ t-1}$, an indicator variable equal to one if the company receives a tax KAM at time *t*, time *t- 1*, or both. H2a predicts a negative coefficient on $TAXKAM\_FIRM_{i,t\ t-1}$, consistent with clients that receive tax KAMs reducing their purchases of APTS. We control for the changes in tax KAM determinants (*ΔTAXAVOIDANCE*, *ΔETR_VOL*, *ΔMATERIALITY*, and *ΔDTA_NET*) and other fee controls, following Carcello and Li (2013) and McGuire et al. (2012). We measure the indicator controls (*HAVEN*, *NOL*, *TAX EXPERT*, *PRIORLOSS*, *BIG_4*, *BUSY, LITIGATE*, and *MERGE_RESTRUCT*) as levels rather than changes due to low inter-year variation in these important factors.

Our second test of H2a examines whether companies that do not receive a tax KAM but would be expected to based on economic fundamentals purchase more APTS. We use the following OLS regression[21]:

[20] All results are robust to clustering standard errors at the year level. Furthermore, given the relatively few observations per firm cluster (2.8 on average) and the variation in the number of observations between firm clusters, we confirm our results are robust to employing nonclustered heteroskedasticity robust standard errors (untabulated) (Cameron and Miller 2015; Omer 2020).

[21] We considered using a subset of firms with tax restatements to identify those with a high probability of receiving tax KAMs that do not receive them but found that the sample size (53) was too small to allow valid inference.

$$LN\_APTS_{it} = \beta_0 + \beta_1 EXPECT\_TAXKAM_{i,t} + X_{i,t} + \mu_{ind} + \tau_t + \varepsilon_{i,t}, \tag{3}$$

where *LN_APTS* is the log of APTS. Following Burke et al. (2023), we measure *EXPECT_TAXKAM*, the expected probability that a company will receive a tax KAM, using the fitted value of *TAXKAM* from the expectations Model (5) in Appendix A.[22] $X_{i,t}$ is a the vector of control variables from Model (2), with continuous variables measured as levels instead of changes to reflect the levels measurement of the dependent variable.

We examine the consequences of tax KAMs to companies' *future* purchases of APTS (H2b) using the following OLS regression:

$$\Delta APTS_{i,t,t+1} = \beta_0 + \beta_1 TAXKAM\_FIRM_{i,tt-1} + \beta_2 RESOLVE\_TAXKAM_{i,t} + \Delta X_{i,t-1,t/}\Delta X_{i,t,t+1} + \mu_{ind} + \tau_t + \varepsilon_{i,t}. \tag{4}$$

$RESOLVE\_TAXKAM_{i,t}$ is an indicator variable equal to one if the company received a tax KAM at time *t - 1* but does not receive one at time *t*. Like many other financial statement disclosures (Brown and Tucker 2011; Kravet and Muslu 2013), tax KAMs are sticky: once a company receives a tax KAM, it is likely to continue to do so.[23] Accordingly, our sample includes 91 observations of companies that resolve tax KAMs.[24] If knowledge spillovers from APTS engagements reduce the complexity of auditing the tax account or raise a self-review threat to auditor independence, we will find a positive association between resolving a tax KAM at time *t* and an increase in concurrent purchases of APTS at time *t* (Model (2)). If APTS resolve tax KAMs at time *t* through self-interest threats to auditor independence, we will not see an increase in these services until the future year *t + 1*.

## 4 Results

### 4.1 Descriptive statistics

We present descriptive statistics for our sample in Table 3. Panel A reports summary statistics for samples used in tests of H1 and H2. The average book

[22] We exclude the *LN_APTS* control from Model (A.1) before estimating the expected value of *TAX_KAM* to avoid inducing mechanical relations with the dependent variables in Model (3). We obtain expected values from the expectations Model (A.1) using the full sample and include only those firms that never receive tax KAMs in the examination of unexpectedly low levels of tax KAM disclosure in Model (3).

[23] Uncertain tax positions and multijurisdictional complexity KAMs are the most persistent tax KAMs, recurring about 70 percent of the time (untabulated). Deferred tax asset KAMs (26 percent) and other tax KAMs (6 percent) are less persistent (untabulated). KAMs on subjects other than taxes (e.g., revenue, inventory) are likewise persistent, except for those relating to nonrecurring economic events, such as business combinations, impairments, and accounting policy changes (untabulated).

[24] We explored the use of a Granger casual model as a formalized approach to distinguish between determinants and consequences. Unfortunately, our time series data does not have sufficient observations per firm to reliably perform such analyses. Specifically, including two lags of each variable requires eight firm-year observations per firm (Sajwan and Chetty 2018).

**Table 3** Descriptive Statistics

| Panel A: Valuation Sample and Auditor-Provided Tax Services Sample | | | | | | |
|---|---|---|---|---|---|---|
| **Variable** | ***N*** | **Mean** | **Std. Dev.** | **P25** | **Median** | **P75** |
| TOBIN'S_Q | 3063 | 1.591 | 1.275 | 0.935 | 1.121 | 1.782 |
| AVOID | 3063 | –0.198 | 0.265 | –0.244 | –0.156 | 0 |
| DTA_NET | 3063 | –0.006 | 0.027 | –0.01 | 0 | 0.001 |
| ETR_VOL | 3063 | 0.125 | 0.188 | 0.006 | 0.031 | 0.132 |
| MATERIALITY | 3063 | 0.011 | 0.019 | 0 | 0.004 | 0.019 |
| HAVEN | 3063 | 0.11 | 0.313 | 0 | 0 | 0 |
| FOREIGN | 3063 | 33.345 | 39.602 | 0 | 4.023 | 75.106 |
| PRIORLOSS | 3063 | 0.521 | 0.5 | 0 | 1 | 1 |
| NOL | 3063 | 0.106 | 0.307 | 0 | 0 | 0 |
| LN_SALES | 3063 | 5.503 | 2.508 | 3.615 | 5.758 | 7.32 |
| SALES_GROWTH | 3063 | 0.167 | 1.111 | –0.099 | 0.215 | 0.182 |
| RETURN_VOL | 3063 | 0.073 | 0.043 | 0.046 | 0.065 | 0.087 |
| LEV | 3063 | 0.182 | 0.204 | 0 | 0.129 | 0.285 |
| INTANGIBLE | 3063 | 0.161 | 0.245 | 0 | 0.021 | 0.268 |
| FCF | 3063 | 0.033 | 0.093 | 0.003 | 0.027 | 0.063 |
| R&D | 3063 | 0.007 | 0.03 | 0 | 0 | 0 |
| CAPEX | 3063 | 0.032 | 0.049 | 0 | 0.014 | 0.046 |
| ACCRUALS | 3063 | –0.003 | 0.11 | –0.056 | –0.014 | 0.039 |
| VOLUME_VOL | 3063 | 7.834 | 1.844 | 6.679 | 7.976 | 9.165 |
| NUM_KAMS | 3063 | 3.035 | 1.437 | 2 | 3 | 4 |
| ΔLN_APTS | 2846 | –1.087 | 3.944 | –0.124 | 0 | 0 |
| %ΔAPTS | 2846 | –0.121 | 0.445 | –0.113 | 0 | 0 |
| TAXKAM_FIRM | 2846 | 0.215 | 0.411 | 0 | 0 | 0 |
| RESOLVE_TAXKAM | 2846 | 0.037 | 0.189 | 0 | 0 | 0 |
| ΔAUDITOR | 2846 | 0.094 | 0.292 | 0 | 0 | 0 |

**Table 3** (continued)

| | | | | | | |
|---|---|---|---|---|---|---|
| BUSY | 2846 | 0.697 | 0.46 | 0 | 1 | 1 |
| INVREC | 2846 | 0.182 | 0.221 | 0.007 | 0.097 | 0.287 |
| MB | 2846 | 2.393 | 3.334 | 0.903 | 1.215 | 2.821 |
| LITIGATE | 2846 | 0.109 | 0.312 | 0 | 0 | 0 |
| TAX_EXPERT | 2846 | 0.223 | 0.417 | 0 | 0 | 0 |
| LN_APTS | 2113 | 2.708 | 4.622 | 0 | 0 | 7.857 |
| EXPECT_TAXKAM | 2113 | 0.08 | 0.133 | 0.003 | 0.015 | 0.098 |

**Panel B: Valuation Sample Companies with and without Tax KAMs**

| | With Tax KAM | | | | Without Tax KAM | | | | Diff. in Means |
|---|---|---|---|---|---|---|---|---|---|
| **Variable** | ***N*** | **Mean** | **Median** | **Std. Dev** | ***N*** | **Mean** | **Median** | **Std. Dev** | ***p*-value** |
| TOBIN'S_Q | 564 | 1.775 | 1.534 | 0.857 | 2499 | 1.493 | 1.035 | 1.289 | 0.000*** |
| AVOID | 564 | –0.259 | –0.219 | 0.254 | 2499 | –0.190 | –0.115 | 0.275 | 0.000*** |
| DTA_NET | 564 | –0.006 | –0.006 | 0.037 | 2499 | –0.005 | 0 | 0.025 | 0.196 |
| ETR_VOL | 564 | 0.143 | 0.074 | 0.166 | 2499 | 0.129 | 0.025 | 0.198 | 0.106 |
| MATERIALITY | 564 | 0.014 | 0.013 | 0.021 | 2499 | 0.009 | 0.002 | 0.018 | 0.000*** |
| HAVEN | 564 | 0.087 | 0 | 0.282 | 2499 | 0.125 | 0 | 0.331 | 0.007*** |
| FOREIGN | 564 | 66.283 | 82.187 | 34.262 | 2499 | 25.370 | 0 | 36.491 | 0.000*** |
| PRIORLOSS | 564 | 0.455 | 0 | 0.498 | 2499 | 0.560 | 1 | 0.496 | 0.000*** |
| NOL | 564 | 0.146 | 0 | 0.353 | 2499 | 0.105 | 0 | 0.306 | 0.003*** |
| LN_SALES | 564 | 7.608 | 7.560 | 1.624 | 2499 | 5.090 | 5.142 | 2.424 | 0.000*** |
| SALES_GROWTH | 564 | 0.067 | 0.003 | 0.705 | 2499 | 0.191 | 0.030 | 1.213 | 0.013** |
| RETURN_VOL | 564 | 0.085 | 0.075 | 0.042 | 2499 | 0.073 | 0.063 | 0.049 | 0.000*** |
| LEV | 564 | 0.262 | 0.249 | 0.193 | 2499 | 0.167 | 0.104 | 0.206 | 0.000*** |
| INTANGIBLE | 564 | 0.295 | 0.273 | 0.258 | 2499 | 0.132 | 0.003 | 0.236 | 0.000*** |
| FCF | 564 | 0.048 | 0.045 | 0.062 | 2499 | 0.028 | 0.021 | 0.095 | 0.000*** |

**Table 3** (continued)

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| R&D | 564 | 0.012 | 0 | 0.029 | 2499 | 0.005 | 0 | 0.029 | 0.000*** |
| CAPEX | 564 | 0.051 | 0.043 | 0.047 | 2499 | 0.028 | 0.004 | 0.048 | 0.000*** |
| ACCRUALS | 564 | –0.050 | –0.042 | 0.080 | 2499 | 0.005 | –0.005 | 0.113 | 0.000*** |
| VOLUME_VOL | 564 | 9.041 | 9.066 | 1.198 | 2499 | 7.562 | 7.614 | 1.855 | 0.000*** |

**Panel C: Valuation Sample Companies with Tax KAM versus with YHAT > 0.75**

| | **With Tax KAM** | | | | **YHAT > 0.75** | | | | **Diff. in Means** |
|---|---|---|---|---|---|---|---|---|---|
| **Variable** | ***N*** | **Mean** | **Median** | **Std. Dev** | ***N*** | **Mean** | **Median** | **Std. Dev** | ***p*-value** |
| TOBIN'S_Q | 564 | 1.775 | 1.534 | 0.857 | 421 | 1.728 | 1.366 | 1.026 | 0.416 |
| AVOID | 564 | –0.259 | –0.219 | 0.254 | 421 | –0.289 | –0.215 | 0.263 | 0.977 |
| DTA_NET | 564 | –0.006 | –0.006 | 0.037 | 421 | –0.008 | –0.002 | 0.031 | 0.410 |
| ETR_VOL | 564 | 0.143 | 0.074 | 0.166 | 421 | 0.149 | 0.071 | 0.175 | 0.575 |
| MATERIALITY | 564 | 0.014 | 0.013 | 0.021 | 421 | 0.013 | 0.008 | 0.020 | 0.283 |
| HAVEN | 564 | 0.087 | 0 | 0.282 | 421 | 0.111 | 0 | 0.314 | 0.188 |
| FOREIGN | 564 | 66.283 | 82.187 | 34.262 | 421 | 66.398 | 75.938 | 31.277 | 0.955 |
| PRIORLOSS | 564 | 0.455 | 0 | 0.498 | 421 | 0.483 | 0 | 0.500 | 0.354 |
| NOL | 564 | 0.146 | 0 | 0.353 | 421 | 0.169 | 0 | 0.375 | 0.311 |
| LN_SALES | 564 | 7.608 | 7.560 | 1.624 | 421 | 7.474 | 7.474 | 1.840 | 0.203 |
| SALES_GROWTH | 564 | 0.067 | 0.003 | 0.705 | 421 | 0.108 | 0.030 | 0.529 | 0.300 |
| RETURN_VOL | 564 | 0.085 | 0.075 | 0.042 | 421 | 0.090 | 0.075 | 0.049 | 0.050** |
| LEV | 564 | 0.262 | 0.249 | 0.193 | 421 | 0.247 | 0.224 | 0.213 | 0.223 |
| INTANGIBLE | 564 | 0.295 | 0.273 | 0.258 | 421 | 0.272 | 0.206 | 0.297 | 0.173 |
| FCF | 564 | 0.048 | 0.045 | 0.062 | 421 | 0.043 | 0.041 | 0.066 | 0.204 |
| R&D | 564 | 0.012 | 0 | 0.029 | 421 | 0.009 | 0 | 0.024 | 0.034** |
| CAPEX | 564 | 0.051 | 0.043 | 0.047 | 421 | 0.045 | 0.033 | 0.044 | 0.033** |
| ACCRUALS | 564 | –0.050 | –0.042 | 0.080 | 421 | –0.048 | –0.037 | 0.084 | 0.660 |
| VOLUME_VOL | 564 | 9.041 | 9.066 | 1.198 | 421 | 8.975 | 9.009 | 1.581 | 0.457 |

effective tax rate for the full sample is 19.8 percent. Each company receives, on average, about three KAMs. Panel B reports characteristics of sample companies from tests of H1 that do and do not receive tax KAMs, revealing that these groups differ on an array of characteristics. Specifically, tax KAM companies have greater sales, derive more of their revenue from foreign operations, and are more likely to have net operating losses. Our additional descriptive statistics (see Appendix A) show that audit fees and APTS are higher for tax KAM companies. The volatility of effective tax rates and the net balance of deferred tax assets and liabilities do not differ significantly across the two subsamples. Panel C compares tax KAM observations with companies that are predicted to receive a tax KAM based on our expectations model (Appendix A) but do not. Our measures of firm value and tax characteristics do not significantly differ across these samples. Additionally, all of our control variables, with the exception of *RETURN_VOL*, *R&D*, and *CAPEX*, are statistically similar across groups, suggesting that our expectations model reasonably captures the company characteristics tied to tax KAM issuance.[25]

### 4.2 Consequences of tax KAM disclosure to the valuation of companies' tax attributes

Table 4 presents the results of testing H1, estimating Model (1) for nontax-KAM, tax-KAM, and predicted-tax-KAM companies. Panel A compares the valuation of tax attributes for tax-KAM and nontax-KAM companies. Among nontax-KAM companies (Columns 1–3), both tax avoidance and deferred tax assets are positively and significantly associated with *TOBIN'S Q*, consistent with the market positively valuing the expected positive future cash flows these tax attributes reflect ($p \leq 0.01$). However, we find no relation between tax avoidance or deferred tax assets and company value for tax KAM companies (Columns 4–6). In tests of differences in the coefficients on AVOID and NET_DTA/DTL across these two samples, we find that the market discounts the tax avoidance (deferred tax assets) of tax-KAM companies compared to that of nontax-KAM companies ($p \leq 0.05$ and $p \leq 0.01$, respectively).[26] Together, these findings imply the positive valuation of tax avoidance and deferred tax assets are fully attenuated for tax-KAM companies, consistent with H1a and b.[27]

To address the possibility that variation in the underlying characteristics of tax-KAM and nontax-KAM companies, as opposed to tax KAMs, drive our results, we examine the tax attribute valuation of companies expected to receive a tax KAM but

[25] We include tax KAM determinant variables (see Appendix A) in subsequent analyses to correct for any remaining imbalance (Shipman et al. 2017).

[26] We cannot rule out that the differences in coefficients are due to the lower power of the test within the subsample of tax KAM firms.

[27] To ensure that multicollinearity among the independent variables is not an issue and is not driving the lack of significant coefficients on our variables of interest in the tax KAM sample, we examine variance inflation factors, which we find to be under 3.5 for all variables in each sample.

**Table 4** Valuation of Tax Avoidance and Deferred Tax Assets

**Panel A: No Tax KAM versus Tax KAM**

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | NO TAX KAM | | | TAX KAM | | |
| VARIABLES | TOBIN'S Q | TOBIN'S Q | TOBIN'S Q | TOBIN'S Q | TOBIN'S Q | TOBIN'S Q |
| AVOID | 0.397*** | | 0.343*** | − 0.036 | | − 0.038 |
| | (2.880) | | (2.877) | (− 0.234) | | (− 0.251) |
| DTA_NET | | 10.701*** | 10.559*** | | 1.087 | 1.089 |
| | | (3.297) | (3.290) | | (0.800) | (0.800) |
| ETR_VOL | − 0.064 | − 0.313** | − 0.102 | − 0.086 | − 0.093 | − 0.115 |
| | (− 0.436) | (− 2.162) | (− 0.726) | (− 0.381) | (− 0.426) | (− 0.510) |
| HAVEN | − 0.122 | − 0.101 | − 0.111 | 0.168 | 0.164 | 0.162 |
| | (− 1.355) | (− 1.185) | (− 1.302) | (1.193) | (1.118) | (1.111) |
| FOREIGN | − 0.001 | − 0.001 | − 0.001 | 0.001 | 0.001 | 0.001 |
| | (− 0.324) | (− 0.364) | (− 0.318) | (0.594) | (0.557) | (0.559) |
| PRIORLOSS | − 0.123** | − 0.129** | − 0.122** | 0.055 | 0.061 | 0.062 |
| | (− 2.298) | (− 2.461) | (− 2.306) | (0.443) | (0.490) | (0.502) |
| MATERIALITY | 40.727*** | 39.710*** | 40.574*** | 11.153*** | 11.150*** | 10.982*** |
| | (5.051) | (5.443) | (5.491) | (3.229) | (3.387) | (3.187) |
| NOL | 0.388*** | 0.458*** | 0.375*** | 0.192* | 0.180* | 0.189* |
| | (3.374) | (4.275) | (3.578) | (1.932) | (1.921) | (1.886) |
| SALES | − 0.122*** | − 0.126*** | − 0.120*** | − 0.006 | − 0.006 | − 0.006 |
| | (− 2.889) | (− 3.071) | (− 3.013) | (− 0.137) | (− 0.149) | (− 0.143) |
| SALES_GROWTH | 0.039 | 0.039* | 0.038* | − 0.040 | − 0.041 | − 0.040 |
| | (1.639) | (1.685) | (1.664) | (− 1.233) | (− 1.267) | (− 1.257) |
| RETURN_VOL | 0.957 | 0.514 | 0.537 | − 2.677** | − 2.945** | − 2.935** |
| | (1.048) | (0.531) | (0.562) | (− 2.226) | (− 2.385) | (− 2.382) |
| LEVERAGE | 0.606 | 0.645 | 0.652 | 0.021 | 0.016 | 0.019 |
| | (1.422) | (1.531) | (1.554) | (0.090) | (0.069) | (0.083) |
| INTANGIBLE | 0.101 | 0.365 | 0.368 | 0.058 | 0.095 | 0.098 |
| | (0.326) | (1.397) | (1.409) | (0.275) | (0.445) | (0.460) |
| FCF | 0.737 | 0.644 | 0.519 | 7.224*** | 7.131*** | 7.181*** |
| | (0.551) | (0.489) | (0.421) | (6.573) | (6.630) | (6.488) |
| R&D | 9.630*** | 8.972*** | 9.076*** | 1.418 | 1.148 | 1.162 |
| | (3.684) | (3.496) | (3.557) | (1.189) | (0.920) | (0.932) |
| CAPEX | 0.664 | 1.092 | 0.902 | 5.275*** | 5.233*** | 5.284*** |
| | (0.368) | (0.661) | (0.546) | (4.311) | (4.419) | (4.342) |
| ACCRUALS | − 1.323** | − 1.000** | − 1.264*** | 1.404*** | 1.352*** | 1.388*** |
| | (− 2.448) | (− 2.314) | (− 2.656) | (2.659) | (2.693) | (2.648) |
| VOLUME_VOL | 0.096** | 0.099** | 0.096** | − 0.058 | − 0.055 | − 0.056 |
| | (2.265) | (2.437) | (2.409) | (− 1.183) | (− 1.113) | (− 1.139) |
| CONSTANT | 1.343*** | 1.390*** | 0.946*** | 1.892*** | 1.907*** | 1.902*** |
| Year and Industry FE | YES | YES | YES | YES | YES | YES |
| Observations | 2499 | 2499 | 2499 | 564 | 564 | 564 |
| R-squared | 0.527 | 0.554 | 0.556 | 0.642 | 0.643 | 0.643 |

**Table 4** (continued)

Tests of Differences Between Models

| Column (3) = Column (6) | Difference | Chi2 |
|---|---|---|
| AVOID | 0.381** | 4.34 |
| DTA_NET | 9.470*** | 7.83 |

**Panel B: Tax KAM versus YHAT > 0.75**

| | **(1)** | **(2)** | **(3)** | **(4)** | **(5)** | **(6)** |
|---|---|---|---|---|---|---|
| | **YHAT > 0.75** | | | **TAX KAM** | | |
| **VARIABLES** | **TOBIN'S Q** | **TOBIN'S Q** | **TOBIN'S Q** | **TOBIN'S Q** | **TOBIN'S Q** | **TOBIN'S Q** |
| AVOID | 0.145 | | 0.161 | − 0.036 | | − 0.038 |
| | (0.884) | | (1.010) | (− 0.234) | | (− 0.251) |
| DTA_NET | | 5.783*** | 5.811*** | | 1.087 | 1.089 |
| | | (3.026) | (3.028) | | (0.800) | (0.800) |
| ETR_VOL | − 0.073 | − 0.231 | − 0.163 | − 0.086 | − 0.093 | − 0.115 |
| | (− 0.270) | (− 0.949) | (− 0.610) | (− 0.381) | (− 0.426) | (− 0.510) |
| HAVEN | − 0.027 | 0.018 | 0.024 | 0.168 | 0.164 | 0.162 |
| | (− 0.191) | (0.125) | (0.166) | (1.193) | (1.118) | (1.111) |
| FOREIGN | 0.003 | 0.005** | 0.005** | 0.001 | 0.001 | 0.001 |
| | (1.551) | (2.150) | (2.140) | (0.594) | (0.557) | (0.559) |
| PRIORLOSS | − 0.119 | − 0.102 | − 0.093 | 0.055 | 0.061 | 0.062 |
| | (− 1.228) | (− 1.052) | (− 0.953) | (0.443) | (0.490) | (0.502) |
| MATERIALITY | 14.566*** | 15.152*** | 15.754*** | 11.153*** | 11.150*** | 10.982*** |
| | (2.946) | (3.176) | (3.295) | (3.229) | (3.387) | (3.187) |
| NOL | 0.110 | 0.165 | 0.110 | 0.192* | 0.180* | 0.189* |
| | (0.805) | (1.418) | (0.814) | (1.932) | (1.921) | (1.886) |
| SALES | − 0.015 | 0.018 | 0.021 | − 0.006 | − 0.006 | − 0.006 |
| | (− 0.295) | (0.356) | (0.405) | (− 0.137) | (− 0.149) | (− 0.143) |
| SALES_GROWTH | 0.069* | 0.043 | 0.046 | − 0.040 | − 0.041 | − 0.040 |
| | (1.914) | (1.302) | (1.378) | (− 1.233) | (− 1.267) | (− 1.257) |
| RETURN_VOL | − 0.925 | − 0.880 | − 1.074 | − 2.677** | − 2.945** | − 2.935** |
| | (− 0.482) | (− 0.450) | (− 0.570) | (− 2.226) | (− 2.385) | (− 2.382) |
| LEVERAGE | − 0.229 | − 0.220 | − 0.192 | 0.021 | 0.016 | 0.019 |
| | (− 0.889) | (− 0.896) | (− 0.782) | (0.090) | (0.069) | (0.083) |
| INTANGIBLE | − 0.247 | 0.009 | − 0.006 | 0.058 | 0.095 | 0.098 |
| | (− 0.870) | (0.033) | (− 0.022) | (0.275) | (0.445) | (0.460) |
| FCF | 4.413*** | 4.262*** | 4.031*** | 7.224*** | 7.131*** | 7.181*** |
| | (3.427) | (3.616) | (3.301) | (6.573) | (6.630) | (6.488) |
| R&D | 3.747 | 3.086 | 3.141 | 1.418 | 1.148 | 1.162 |
| | (1.246) | (0.995) | (1.009) | (1.189) | (0.920) | (0.932) |
| CAPEX | 3.741** | 4.012** | 3.848** | 5.275*** | 5.233*** | 5.284*** |
| | (2.064) | (2.330) | (2.306) | (4.311) | (4.419) | (4.342) |
| ACCRUALS | 1.101* | 1.254** | 1.063* | 1.404*** | 1.352*** | 1.388*** |
| | (1.819) | (2.320) | (1.862) | (2.659) | (2.693) | (2.648) |

**Table 4** (continued)

| | | | | | | |
|---|---|---|---|---|---|---|
| VOLUME_VOL | – 0.033<br>(– 0.650) | – 0.031<br>(– 0.638) | – 0.031<br>(– 0.646) | – 0.058<br>(– 1.183) | – 0.055<br>(– 1.113) | – 0.056<br>(– 1.139) |
| CONSTANT | 1.370** | 1.079* | 1.092* | 1.892*** | 1.907*** | 1.902*** |
| | (2.135) | (1.829) | (1.877) | (4.764) | (4.760) | (4.747) |
| Year and Industry FE | YES | YES | YES | YES | YES | YES |
| Observations | 418 | 418 | 418 | 564 | 564 | 564 |
| R-squared | 0.689 | 0.703 | 0.703 | 0.642 | 0.643 | 0.643 |
| Tests of Differences Between Models | | | | | | |
| Column (3) = Column (6) | Difference | Chi2 | | | | |
| AVOID | 0.199 | 1.03 | | | | |
| DTA_NET | 4.722** | 5.42 | | | | |

This table reports the results of OLS regression of company value (TOBIN'S Q) on companies' tax attributes (AVOID and DTA_NET) and a set of control variables. Panel A reports differences in valuation effects of tax attributes for firms that do not receive a tax KAM (columns 1–3) and those that do (columns 4–6). Panel B reports differences in valuation effects of tax attributes for firms that receive a tax KAM (columns 1–3) and firms predicted to receive a tax KAM (yhat > 0.75) but do not. In all regressions, standard errors are clustered by company. Two-tailed t-statistics are presented in brackets, with Chi2 statistics presented in tests of differences in coefficients. The symbols ***, **, and * represent two-tailed p-values significant at the 0.01, 0.05 and 0.10 levels, respectively

that do not. To identify these companies, we construct an expectations model for tax KAM issuance, which we describe and test in Appendix A. Consistent with our expectations, we find that several of our measures of tax complexity are positively associated with tax KAM issuance, including tax avoidance (*AVOID*), deferred tax asset balances (*DTA_NET*), and the extent of foreign operations (*FOREIGN*). We do not find a significant association between APTS or nontax non-audit services and tax KAM issuance, providing no initial evidence of an APTS effect. Using the predicted value of tax KAM from Appendix A for our sample of nontax-KAM companies, we designate observations with predicted values greater than 0.75 as having unexpectedly low levels of tax KAM disclosure. Table 4 Panel B Columns 1–3 present the results of estimating Model (1) for these companies. We find that the market positively values the tax attributes of companies with unexpectedly low levels of tax KAM disclosure, though only the coefficient on deferred tax assets is significant ($p < 0.01$). In tests of differences between tax KAM and predicted tax KAM companies, we find that the differential valuation of deferred tax assets among the two subsamples is significantly different from zero. Overall, these findings suggest that the valuation effects of tax KAMs are incremental to other company-specific factors, supporting H1.

### 4.3 Consequences of tax KAMs to auditor-provided tax services

#### 4.3.1 Concurrent association between tax KAMs and auditor-provided tax services

We present results for H2a in Table 5. Panel A shows that companies with tax KAMs (*TAXKAM_FIRM)* decrease their fees for APTS by 32 percent, or $63,125

US dollars, in the year of KAM issuance (Column 1, $\beta = -0.317$, $p \leq 0.10$).[28] These findings are consistent with either the client economically punishing the auditor for issuing the tax KAM, the auditor punishing the client for reduced APTS by disclosing a tax KAM, or a knowledge spillover effect where the lack of APTS increases the difficulty of auditing the tax function. We also find that this result is robust to a subsample of only those companies that receive a tax KAM at least once during the sample period (untabulated). Table 5 Panel B shows that companies with unexpectedly low levels of tax KAM disclosures pay higher fees for APTS (Column 1, $\beta = 4.395$, $p \leq 0.05$), consistent with clients purchasing APTS to avoid initial tax KAM disclosure. Column 3 illustrates that these results are robust to examining a subsample of companies that purchase APTS at least once during the sample period. Collectively, these results support H2a, that tax KAMs relate negatively to concurrent purchases of APTS. We next examine whether this association is consistent with knowledge spillovers or economic bond threats to independence.

#### 4.3.2 Consequences of tax KAMs to future auditor-provided tax services

We present results for H2b in Table 5 Panel C. We find that companies that resolve a tax KAM at time *t* increase purchases of APTS by 104 percent, or $207,894 US dollars, from time *t* to *t + 1* (Column 1, $\beta = 1.044$, $p \leq 0.05$).[29] We find no significant contemporaneous associations between resolving a tax KAM and changes in APTS from *t- 1* to *t* (Table 5, Panel A), suggesting that fees for APTS do not increase until the year *after* the tax KAM resolution. This timing supports the explanation that the economic incentive of *future* APTS compromises creates a self-interest threat to the auditor's independence in issuing the KAM. This timing is inconsistent with knowledge spillovers or self-review threats contributing to KAM resolution, as the provision of tax services does not begin until the year *after* the tax KAM's resolution. As KAM reporting in general is not associated with changes in audit fees (Bédard et al. 2019; Gutierrez et al. 2018), these results illustrate a topic-specific effect of tax KAMs on auditors' economic rewards. Column 3 of Table 5 Panel C illustrates that this finding is robust to operationalizing the change in APTS as the percentage change in these services.[30] These results are also robust to the subsample of companies that receive a tax KAM at least once during the sample period (untabulated).

[28] The $63,125 decrease represents a 31.7 percent reduction in the mean tax fees of $199,132 among in-sample companies that purchase APTS.

[29] The $207,894 increase represents a 104.4 percent increase in the mean tax fees of $199,132 among in-sample companies that purchase APTS.

[30] In untabulated analyses, we find this result is robust to measuring the dependent variable as APTS scaled by any of the following: total assets, the square root of total assets, SG&A expense, audit fees, and audit and related fees.

**Table 5** Consequences of Tax KAMs for Auditor-Provided Tax Services

**Panel A: Consequences of Tax KAM Disclosure for Concurrent Auditor-Provided Tax Services Purchases**

| | **(1)** | **(2)** | **(3)** | **(4)** |
|---|---|---|---|---|
| | **$\Delta$LN_APTS$_{i,t-1,t}$** | | **%$\Delta$APTS$_{i,t-1,t}$** | |
| **VARIABLES** | **coef** | **tstat** | **coef** | **tstat** |
| TAXKAM_FIRM$_{i,t,\ t-1}$ | − 0.317* | (− 1.76) | − 0.050** | (− 2.22) |
| RESOLVE_TAXKAM$_{i,t}$ | 0.010 | (0.02) | 0.013 | (0.26) |
| ΔTAXAVOIDANCE | − 0.095 | (− 0.35) | − 0.032 | (− 1.10) |
| ΔETR_VOL | 1.024** | (2.03) | 0.025 | (0.46) |
| HAVEN | 0.973*** | (5.46) | 0.133*** | (5.41) |
| ΔMATERIALITY | − 4.637 | (− 0.65) | − 0.949 | (− 1.04) |
| NOL | − 0.029 | (− 0.10) | 0.017 | (0.54) |
| ΔDTA_NET | 8.067 | (0.84) | − 0.167 | (− 0.16) |
| ΔASSETS | 0.864** | (1.99) | 0.083* | (1.72) |
| ΔPTI | − 0.272 | (− 0.47) | − 0.072 | (− 1.01) |
| ΔLEV | 2.090** | (2.38) | 0.257*** | (2.63) |
| BIG_4 | − 0.227 | (− 1.52) | − 0.025 | (− 1.13) |
| ΔAUDITOR | − 2.177*** | (− 6.26) | − 0.194*** | (− 5.70) |
| BUSY | − 0.378*** | (− 3.44) | − 0.039*** | (− 2.72) |
| ΔNUM_KAMS | − 0.046 | (− 0.54) | − 0.008 | (− 0.97) |
| ΔINVREC | − 2.592** | (− 2.32) | − 0.282*** | (− 2.64) |
| ΔMB | − 0.015 | (− 0.57) | − 0.001 | (− 0.26) |
| ΔSALES | − 0.102 | (− 0.78) | − 0.007 | (− 0.37) |
| LITIGATE | − 0.460* | (− 1.88) | − 0.056 | (− 1.52) |
| ΔFOREIGN | 0.002 | (0.26) | 0.000 | (0.26) |
| KAM_REGPERIOD | − 1.509*** | (− 4.11) | − 0.170*** | (− 3.99) |
| TAX_EXPERT | 0.632*** | (3.64) | 0.072*** | (3.28) |
| ΔFCF | − 0.702 | (− 0.73) | − 0.025 | (− 0.23) |
| MERGE_RESTRUCT | − 0.226 | (− 1.40) | − 0.029 | (− 1.49) |

**Table 5** (continued)

| | | | | |
|---|---|---|---|---|
| PRIORLOSS | 0.235* | (1.94) | 0.017 | (1.11) |
| Constant | 0.372 | (0.92) | − 0.019 | (− 0.38) |
| Observations | 2,846 | | 2,846 | |
| R-squared | 0.089 | | 0.077 | |
| Year FE | Yes | | Yes | |
| Industry FE | Yes | | Yes | |

**Panel B: Consequences of Unexpectedly Low Levels of Tax KAM Disclosure for Auditor-Provided Tax Services**

| | **All Firms that Never Receive a Tax KAM** | | **Never Tax KAM Firms that Purchase Auditor-Provided Tax Services During the Sample Period** | |
|---|---|---|---|---|
| | **(1)** | **(2)** | **(3)** | **(4)** |
| | **LN_APTS** | | **LN_APTS** | |
| **VARIABLES** | **coef** | **tstat** | **coef** | **tstat** |
| EXPECT_TAXKAM | 4.395** | (2.06) | 1.833** | (2.14) |
| TAXAVOIDANCE | 0.472 | (1.24) | − 0.375 | (− 1.14) |
| ETR_VOL | 0.669 | (1.08) | − 0.647 | (− 1.53) |
| DTA_NET | 1.870 | (0.23) | 3.407 | (0.96) |
| HAVEN | 1.141** | (2.09) | 0.604* | (1.93) |
| MATERIALITY | − 0.174 | (− 0.01) | 4.378 | (0.71) |
| FOREIGN | 0.003 | (0.55) | − 0.000 | (− 0.08) |
| NOL | 0.408 | (0.77) | 0.323 | (1.35) |
| ASSETS | − 0.210* | (− 1.70) | 0.183*** | (3.02) |
| PTI | − 1.703 | (− 1.60) | − 0.638 | (− 0.85) |
| LEV | − 0.058 | (− 0.07) | 1.004** | (2.58) |
| BIG_4 | − 0.312 | (− 0.77) | 0.335 | (1.62) |
| ΔAUDITOR | − 1.372*** | (− 5.51) | − 0.087 | (− 0.30) |
| BUSY | 0.546* | (1.77) | 0.081 | (0.42) |
| NUM_KAMS | − 0.191 | (− 1.19) | − 0.019 | (− 0.26) |

**Table 5** (continued)

| | | | | |
|---|---|---|---|---|
| INVREC | 1.544* | (1.95) | 0.285 | (0.85) |
| MB | − 0.031 | (− 0.59) | − 0.007 | (− 0.34) |
| FCF | 2.542 | (1.51) | 1.014 | (1.15) |
| SALES_GROWTH | 0.074 | (1.23) | 0.041 | (1.15) |
| LITIGATE | − 0.124 | (− 0.15) | 0.037 | (0.12) |
| TAX_EXPERT | 1.526*** | (5.29) | 0.348*** | (2.70) |
| MERGE_RESTRUCT | 0.859** | (2.48) | 0.441** | (2.15) |
| KAM_REGPERIOD | − 2.219*** | (− 5.04) | − 0.774*** | (− 3.92) |
| PRIORLOSS | − 0.533* | (− 1.74) | − 0.336** | (− 2.13) |
| Constant | 6.749*** | (4.73) | 9.266*** | (17.65) |
| Observations | 2,113 | | 553 | |
| R-squared | 0.264 | | 0.523 | |
| Year FE | Yes | | Yes | |
| Industry FE | Yes | | Yes | |
| **Panel C: Consequences of Tax KAM Disclosure for Future Auditor-Provided Tax Services Purchases** | | | | |
| | **(1)** | **(2)** | **(3)** | **(4)** |
| | **$\Delta$LN_APTS$_{i,t,t+1}$** | | **%$\Delta$APTS$_{i,t,t+1}$** | |
| **VARIABLES** | **coef** | **tstat** | **coef** | **tstat** |
| TAXKAM_FIRM$_{i,t,t-1}$ | − 0.304* | (− 1.79) | − 0.039* | (− 1.86) |
| RESOLVE_TAXKAM$_{i,t}$ | 1.044** | (2.38) | 0.191*** | (3.59) |
| $\Delta$TAXAVOIDANCE | − 0.109 | (− 0.40) | − 0.035 | (− 1.20) |
| $\Delta$ETR_VOL | 1.005** | (1.99) | 0.021 | (0.39) |
| HAVEN | 0.975*** | (5.48) | 0.133*** | (5.47) |
| $\Delta$MATERIALITY | − 5.072 | (− 0.71) | − 1.056 | (− 1.15) |
| NOL | − 0.035 | (− 0.12) | 0.015 | (0.50) |
| $\Delta$DTA_NET | 8.038 | (0.84) | − 0.192 | (− 0.19) |
| $\Delta$ASSETS | 0.859** | (1.98) | 0.082* | (1.70) |
| $\Delta$PTI | − 0.260 | (− 0.45) | − 0.069 | (− 0.98) |

**Table 5** (continued)

| | | | | |
|---|---|---|---|---|
| ΔLEV | 2.083** | (2.37) | 0.256*** | (2.64) |
| BIG_4 | − 0.237 | (− 1.58) | − 0.028 | (− 1.26) |
| ΔAUDITOR | − 2.179*** | (− 6.29) | − 0.194*** | (− 5.73) |
| BUSY | − 0.381*** | (− 3.46) | − 0.039*** | (− 2.77) |
| ΔNUM_KAMS | − 0.052 | (− 0.61) | − 0.009 | (− 1.09) |
| ΔINVREC | − 2.594** | (− 2.32) | − 0.282*** | (− 2.64) |
| ΔMB | − 0.014 | (− 0.53) | − 0.001 | (− 0.19) |
| ΔSALES | − 0.102 | (− 0.78) | − 0.007 | (− 0.37) |
| LITIGATE | − 0.467* | (− 1.92) | − 0.057 | (− 1.56) |
| ΔFOREIGN | 0.002 | (0.21) | 0.000 | (0.17) |
| KAM_REGPERIOD | − 1.522*** | (− 4.16) | − 0.173*** | (− 4.06) |
| TAX_EXPERT | 0.636*** | (3.68) | 0.072*** | (3.32) |
| ΔFCF | − 0.717 | (− 0.75) | − 0.027 | (− 0.25) |
| MERGE_RESTRUCT | − 0.238 | (− 1.47) | − 0.033 | (− 1.64) |
| PRIORLOSS | 0.241** | (1.99) | 0.018 | (1.21) |
| Constant | 0.349 | (0.88) | − 0.026 | (− 0.51) |
| Observations | 2,846 | | 2,846 | |
| R-squared | 0.090 | | 0.080 | |
| Year FE | Yes | | Yes | |
| Industry FE | Yes | | Yes | |

Panel A: This panel reports the results of estimating Model (2), which examines the association between disclosing tax KAMs (TAXKAM_FIRM) and resolving tax KAMs (RESOLVE_TAXKAM) and concurrent purchases of auditor-provided tax services. We use an ordinary least squares regression with standard errors clustered on firm. The symbols ***, **, and * represent two-tailed p-values significant at the 0.01, 0.05 and 0.10 levels, respectively. Two-tailed t-statistics are presented in brackets

Panel B: This panel reports the results of estimating Model (3), which examines whether unexpectedly low levels of tax KAM disclosure are associated with higher purchases of auditor-provided tax services. We use ordinary least squares regression with standard errors clustered on firm. The symbols ***, **, and * represent two-tailed p-values significant at the 0.01, 0.05 and 0.10 levels, respectively. Two-tailed t-statistics are presented in brackets

Panel C: This panel reports the results of estimating Model (4), which examines the association between disclosing tax KAMs (TAXKAM_FIRM) and resolving tax KAMs (RESOLVE_TAXKAM) and future purchases of auditor-provided tax services. We use an ordinary least squares regression with standard errors clustered on firm. The symbols ***, **, and * represent two-tailed p-values significant at the 0.01, 0.05 and 0.10 levels, respectively. Two-tailed t-statistics are presented in brackets

# 5 Additional analyses

## 5.1 Specific tax KAM topics

Different subcategories of tax KAMs may have different consequences for the valuation of company tax attributes and the incentives to purchase APTS. For example, clients may be more concerned about KAMs that could highlight their uncertain tax positions to regulators or call into question their ability to generate future taxable income to realize deferred tax assets. Conversely, clients may be unconcerned about tax KAMs that reflect the complexity of their normal operations, such as doing business in many foreign jurisdictions and related transfer pricing matters. Further, we predict that tax KAMs referencing the subjective issues of uncertain tax positions and deferred taxes afford the auditor more discretion concerning whether to issue tax KAMs than issues of multijurisdictional complexity, which are inherent to the company's geographic footprint. This interpretation is consistent with the literature showing that accounting for uncertain tax positions and deferred taxes is susceptible to earnings management due to the subjectivity involved in their valuation (Phillips et al. 2003; Schrand and Wong 2003; Blouin and Robinson 2014; Cazier et al. 2015; Gupta et al. 2016). We therefore disaggregate tax KAMs into subcategories based on the issues they discuss.

We begin our disaggregation efforts by considering Audit Analytics' classification of tax KAMs into three subcategories: uncertain tax positions, deferred income taxes, and other income taxes. However, upon manual review of the text of tax KAMs, we found these categorizations unreliable.[31] As a result, we read all tax KAM disclosures, identifying and cataloging four primary and nonmutually exclusive subtypes: (1) uncertain tax positions, (2) realizability of deferred tax assets, (3) multijurisdictional complexity, and (4) other tax KAMs.[32] See Appendix D for a detailed description of our hand-coding process and a correlation table illustrating the material differences in our classification versus Audit Analytics'. For example, there is only a 32 percent overlap in the matters determined to be uncertain tax position KAMs under the two categorization schemes. Panel A of Table 2 lists the number and types of tax KAMs by year.

We first test whether the detailed issues the auditor discusses in the tax KAMs map into.

companies' tax characteristics by estimating an expectations model for each type of tax KAM. Table 6 Panel A illustrates that companies' tax avoidance is the strongest predictor of receiving an uncertain tax position KAM, with a positive coefficient on *AVOID* ($\beta = 0.054$, $p \leq 0.05$) and that *DTA_NET* is positively associated with *DTA_KAM* ($\beta = 0.650$, $p \leq 0.01$) but not with other tax

[31] For example, PwC's 2018 audit report for G4S PLC includes a key audit matter titled "uncertain tax positions and deferred tax assets," which was categorized as "other income taxes" by Audit Analytics. Prior periods included nearly identical KAMs, which were properly coded as "uncertain tax position" KAMs.

[32] As an example of our nonmutually exclusive categorization approach, we classify the KAM presented in Appendix B both as a uncertain tax position KAM and as a complexity KAM because the auditor discusses both specific tax positions and overall complexity creating challenges in the audit.

**Table 6** Specific Tax KAM Topics

| **Panel A: Predicting Specific Tax KAM Topics** | | | | |
|---|---|---|---|---|
| **VARIABLES** | **(1) UTP_KAM** | **(2) DTA_KAM** | **(3) COMP_KAM** | **(4) OTHER_KAM** |
| AVOID | 0.054** | 0.015 | 0.000 | – 0.001 |
| | (2.292) | (1.249) | (0.028) | (– 0.208) |
| ETR_VOL | 0.036 | 0.045* | – 0.084** | 0.003 |
| | (0.897) | (1.848) | (– 2.362) | (0.317) |
| DTA_NET | 0.168 | 0.650*** | 0.299 | – 0.054 |
| | (0.715) | (3.381) | (1.531) | (– 0.960) |
| FOREIGN | 0.001*** | 0.000*** | 0.001*** | – 0.000 |
| | (4.650) | (2.840) | (5.199) | (– 0.989) |
| HAVEN | – 0.016 | 0.004 | 0.033 | – 0.012 |
| | (– 0.614) | (0.198) | (1.320) | (– 1.454) |
| MATERIALITY | – 0.046 | – 0.278 | – 0.691* | 0.118 |
| | (– 0.107) | (– 0.920) | (– 1.767) | (0.707) |
| BIG4 | 0.069** | 0.109*** | 0.055** | – 0.009 |
| | (2.544) | (3.197) | (2.317) | (– 1.286) |
| NOL | – 0.002 | – 0.003 | – 0.017 | |
| | (– 0.111) | (– 0.221) | (– 1.121) | |
| PRIORLOSS | – 0.006 | 0.047*** | 0.003 | 0.001 |
| | (– 0.329) | (3.450) | (0.212) | (0.218) |
| LN_AUDIT FEES | 0.028** | 0.033*** | 0.038*** | 0.004 |
| | (2.435) | (3.659) | (2.902) | (1.306) |
| LN_APTS | – 0.000 | 0.000 | 0.001 | – 0.000 |
| | (– 0.204) | (0.314) | (1.104) | (– 0.782) |
| LN_NONTAXNAS | 0.000 | – 0.000 | 0.002 | 0.000 |
| | (0.052) | (– 0.064) | (1.354) | (0.615) |
| MERGE_RESTRUCT | 0.003 | 0.005 | – 0.003 | 0.004 |
| | (0.203) | (0.433) | (– 0.221) | (0.718) |
| ASSETS | 0.011 | – 0.011* | – 0.000 | 0.000 |
| | (1.292) | (– 1.735) | (– 0.041) | (0.240) |
| PTI | 0.010 | 0.082** | – 0.012 | 0.041 |
| | (0.139) | (2.042) | (– 0.206) | (1.285) |
| FCF | 0.052 | – 0.053 | 0.240*** | 0.043 |
| | (0.509) | (– 1.052) | (2.665) | (1.012) |
| NUM_KAMS | 0.025*** | 0.011*** | 0.011*** | 0.003** |
| | (4.376) | (2.698) | (2.720) | (2.125) |
| KAM_REGPERIOD | 0.017 | 0.004 | 0.032 | 0.097*** |
| | (0.676) | (0.220) | (1.598) | (3.742) |
| Year FE | YES | YES | YES | YES |
| Industry FE | YES | YES | YES | YES |
| Observations | 4,008 | 3,955 | 3,955 | 2,975 |
| Area Under Curve | 0.900 | 0.897 | 0.937 | 0.878 |

**Table 6** (continued)

**Panel B: Specific Tax KAM Topics and Valuation of Company Tax Attributes**

**Valuation of Tax Attributes**

| | **UTP KAMs** | | **DTA KAMs** | |
|---|---|---|---|---|
| **VARIABLES** | **(1)** **No UTP KAM** **TOBIN'S Q** | **(2)** **UTP KAM** **TOBIN'S Q** | **(3)** **No DTA KAM** **TOBIN'S Q** | **(4)** **DTA KAM** **TOBIN'S Q** |
| AVOID | 0.348*** | − 0.064 | 0.365*** | 0.283 |
| | (2.837) | (− 0.389) | (2.783) | (1.487) |
| DTA_NET | 9.664*** | 0.130 | 7.728*** | − 2.627 |
| | (3.236) | (0.086) | (3.010) | (− 1.015) |
| Controls | YES | YES | YES | YES |
| Year and Industry FE | YES | YES | YES | YES |
| Observations | 2,621 | 442 | 2,877 | 186 |
| R-squared | 0.547 | 0.705 | 0.528 | 0.743 |

**Tests of Differences Between Models**

| | Difference | Chi2 | p-value |
|---|---|---|---|
| **UTP KAMs:** Column (1) = Column (2) | | | |
| AVOID | 0.413** | 4.54 | (0.033) |
| DTA_NET | 9.534*** | 8.81 | (0.003) |
| **DTA KAMs:** Column (3) = Column (4) | | | |
| AVOID | 0.082 | 0.17 | (0.682) |
| DTA_NET | 10.355*** | 9.86 | (0.002) |
| **UTP versus DTA KAMs:** Column (2) = Column (4) | | | |
| AVOID | − 0.347* | 2.82 | (0.093) |
| DTA_NET | 2.757 | 1.22 | (0.270) |

**Panel C: Unexpectedly Low Levels of Specific Tax KAM Topic Disclosures and Auditor-Provided Services**

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| **VARIABLES** | **LN_APTS** | **LN_APTS** | **LN_APTS** | **LN_APTS** |
| EXPECT_UTPKAM | 5.233* | | | |
| | (1.95) | | | |
| EXPECT_DTAKAM | | 9.084** | | |
| | | (2.25) | | |
| EXPECT_COMPKAM | | | 5.883** | |
| | | | (2.40) | |
| EXPECT_OTHERKAM | | | | 5.496 |
| | | | | (0.49) |

**Table 6** (continued)

| | | | | |
|---|---|---|---|---|
| TAXAVOIDANCE | 0.476 | 0.604 | 0.615* | 1.220*** |
| | (1.24) | (1.65) | (1.65) | (3.10) |
| ETR_VOL | 0.613 | 0.437 | 0.675 | 1.336** |
| | (0.98) | (0.71) | (1.10) | (2.04) |
| DTA_NET | 4.468 | − 2.158 | 3.426 | − 5.367 |
| | (0.56) | (− 0.24) | (0.42) | (− 0.54) |
| HAVEN | 1.193** | 1.017* | 0.954* | 0.839 |
| | (2.19) | (1.89) | (1.77) | (1.54) |
| MATERIALITY | 0.205 | 0.570 | 2.526 | 1.488 |
| | (0.02) | (0.05) | (0.20) | (0.07) |
| NOL | 0.423 | 0.444 | 0.619 | |
| | (0.80) | (0.81) | (1.17) | |
| Observations | 2,113 | 2,081 | 2,081 | 1,634 |
| R-squared | 0.264 | 0.268 | 0.268 | 0.237 |
| Year FE | Yes | Yes | Yes | Yes |
| Industry FE | Yes | Yes | Yes | Yes |

**Panel D: Consequences of Resolving Specific Tax KAM Topics for Auditor-Provided Tax Services**

| | ***Concurrent* Change (t- 1 to t)** | | ***Future* Change (t to t + 1)** | |
|---|---|---|---|---|
| | **(1)** | (2) | (3) | (4) |
| **VARIABLES** | **$\Delta$LN_APTS$_{i,t-1,t}$** | **%$\Delta$APTS$_{i,t-1,t}$** | **$\Delta$LN_APTS$_{i,t,t+1}$** | **%$\Delta$APTS$_{i,t,t+1}$** |
| UTPKAM_FIRM$_{i,t,\ t-1}$ | − 0.122 | − 0.036 | − 0.204 | − 0.049** |
| | (− 0.65) | (− 1.49) | (− 1.11) | (− 2.20) |
| RESOLVE_UTPKAM$_{i,t}$ | − 0.250 | − 0.037 | 1.148** | 0.163*** |
| | (− 0.42) | (− 0.64) | (1.99) | (2.82) |
| DTAKAM_FIRM$_{i,t,\ t-1}$ | − 0.487 | − 0.038 | − 0.224 | 0.001 |
| | (− 1.65) | (− 1.07) | (− 0.84) | (0.03) |
| RESOLVE_DTAKAM$_{i,t}$ | 0.257 | 0.040 | 1.535*** | 0.214*** |
| | (0.39) | (0.59) | (3.08) | (2.85) |
| COMPKAM_FIRM$_{i,t,\ t-1}$ | − 0.003 | − 0.030 | − 0.329 | − 0.047* |
| | (− 0.01) | (− 1.16) | (− 1.49) | (− 1.82) |
| RESOLVE_COMPKAM$_{i,t}$ | − 0.788 | − 0.048 | 0.682 | 0.102 |
| | (− 0.99) | (− 0.62) | (0.78) | (0.92) |
| OTHERKAM_FIRM$_{i,t,\ t-1}$ | − 0.959 | − 0.108 | − 1.117 | − 0.110 |
| | (− 1.40) | (− 1.46) | (− 1.32) | (− 1.24) |
| RESOLVE_OTHERKAM$_{i,t}$ | 0.411 | 0.064 | 1.751** | 0.168* |
| | (0.30) | (0.57) | (2.06) | (1.88) |

**Table 6** (continued)

| | | | | |
|---|---|---|---|---|
| Observations | 2,846 | 2,846 | 2,846 | 2,846 |
| R-squared | 0.088 | 0.077 | 0.090 | 0.079 |
| Year FE | Yes | Yes | Yes | Yes |
| Industry FE | Yes | Yes | Yes | Yes |
| Controls | Yes | Yes | Yes | Yes |

Panel A: This panel presents the results of estimating Model (5), which predicts the likelihood that a company will receive a tax KAM, for each of the specific tax KAM topics described in Appendix D: uncertain tax positions, deferred tax assets, multijurisdictional complexity, and other. We use a probit regression with standard errors clustered by firm. Coefficients represent average marginal effects. Two-tailed t-statistics are presented in brackets. The symbols ***, **, and * represent two-tailed p-values significant at the 0.01, 0.05 and 0.10 levels, respectively

Panel B: This panel reports the valuation of tax attributes across tax KAM subtypes obtained by estimating Model (1) using an OLS regression with standard errors clustered by firm. First, we present the valuation of these attributes for firms receiving and not receiving an uncertain tax position-related tax KAM and for firms receiving and not receiving a deferred tax asset-related tax KAM. Then we report differences in coefficients across these subsamples. Untabulated control variables include ETR_VOL, HAVEN, FOREIGN, PRIORLOSS, MATERIALITY, NOL, SALES, SALES_GROWTH, RETURN_VOL, LEVERAGE, INTANGIBLE, FCF, R&D, CAPEX, ACCRUAL, and VOLUME_VOL. Two-tailed t-statistics are presented in brackets (Chi2 statistics presented in tests of differences). The symbols ***, **, and * represent two-tailed p-values significant at the 0.01, 0.05 and 0.10 levels, respectively

Panel C: This panel reports the results of estimating Model (3), which examines whether unexpectedly low levels of tax KAM disclosure are associated with higher purchases of auditor-provided tax services for each specific topic of tax KAM. We use an ordinary least squares regression with standard errors clustered on firm. The symbols ***, **, and * represent two-tailed p-values significant at the 0.01, 0.05 and 0.10 levels, respectively. Two-tailed t-statistics are presented in brackets. Untabulated control variables include ASSETS, PTI, LEV, BIG_4, AUDITOR, BUSY, NUM_KAMS, INVREC, MB, SALES_GROWTH, LITIGATE, FOREIGN, KAM_REGPERIOD, TAX_EXPERT, FCF, MERGE_RESTRUCT, and PRIORLOSS

Panel D: This panel reports the results of examining the association between disclosing specific tax KAM topics (KAM_FIRM) and resolving specific types of tax KAMs (RESOLVE_KAM) on concurrent and next-period purchases of auditor-provided tax services. Columns 1 and 2 present the results of Model (2), which examines concurrent purchases of auditor-provided tax services, and Columns 3 and 4 present the results of Model (4), which examines next-period purchases of auditor-provided tax services, for each specific topic of tax KAM described in Appendix D. We use an ordinary least squares regression with standard errors clustered on company. The symbols ***, **, and * represent two-tailed p-values significant at the 0.01, 0.05 and 0.10 levels, respectively. Two-tailed t-statistics are presented in brackets. Untabulated control variables include ΔASSETS, ΔPTI, ΔLEV, BIG_4, ΔAUDITOR, BUSY, ΔNUM_KAMS, ΔINVREC, ΔMB, ΔSALES, LITIGATE, ΔFOREIGN, KAM_REGPERIOD, TAX_EXPERT, ΔFCF, MERGE_RESTRUCT, and PRIORLOSS

KAM types. We find that multijurisdictional complexity tax KAMs are associated with greater foreign sales ($\beta = 0.001$, $p \leq 0.01$). Our tax-related variables generally do not predict "other" tax KAM issuance, consistent with these KAMs arising from idiosyncratic issues. Collectively, these results show that auditors' tax KAM descriptions closely reflect client economic circumstances and tailor the content of tax KAMs to specific drivers of the companies' tax complexity rather than using generic descriptions across all tax KAMs. Our findings demonstrate the importance of carefully categorizing KAM topics for valid expanded audit reporting research.

#### 5.1.1 Tax KAM subtypes and valuation of companies' tax attributes

We posit that the valuation of company tax KAMs may vary across specific tax KAM subtopics. We therefore re-estimate Model (1) for uncertain tax position and deferred tax asset KAMs, which directly relate to our tax attributes of interest. We present the results in Table 6 Panel B. We find that receiving an uncertain tax position KAM fully attenuates the positive market valuation of *both* tax avoidance and deferred tax assets. We show that receiving a deferred tax asset KAM does not significantly reduce the valuation of companies' tax avoidance but fully attenuates the positive valuation of deferred tax assets (diff. $=10.355$, $p< 0.01$). When comparing uncertain tax position KAMs to deferred tax asset KAMs, we find that companies receiving the former KAMs have lower valued tax avoidance than those receiving the latter ones (diff. $=-0.347$, $p\leq 0.10$). Together, these results suggest the valuation of tax attributes varies across the specific tax information in the tax KAM.

#### 5.1.2 Tax KAM subtypes and auditor-provided tax services

To determine whether the relation between tax KAM issuance and auditor-provided tax services differs across KAM subtypes based on the level of auditor discretion, we re-estimate Models (2), (3), and (4) for tax KAM subtypes in Table 6 Panels C and D. In Table 6 Panel C, we find that unexpectedly low levels of deferred tax asset KAM disclosure are most positively associated with APTS (Column 2, $\beta= 9.084$, $p\leq 0.05$), followed in magnitude by uncertain tax position KAMs (Column 1, $\beta= 5.223$, $p\leq 0.10$) and complexity KAMs (Column 3, $\beta= 5.883$, $p\leq 0.05$). In Table 6 Panel D, we find that resolving uncertain tax position and deferred tax asset KAMs at time $t$ are associated with 115 and 154 percent increases in APTS at $t+1$, respectively (Column 3). We do not find a similar association among companies receiving tax KAMs related to multijurisdictional complexity. Collectively, these results imply that the relation between tax KAM disclosure and APTS is strongest where the auditor has greater discretion to elect not to disclose a KAM and where the client has greater incentive to seek to resolve a tax KAM (for uncertain tax position and deferred tax asset KAMs). This implies that economic bonds, rather than knowledge spillovers, drive the relation between tax KAMs and APTS. We posit that the tax KAM subtypes resolved by knowledge spillovers would not vary with auditor discretion, and, if they did so, the tax KAM would be resolved in the year of the APTS engagement rather than the year *before* the APTS engagement took place.

### 5.2 Controlling for tax footnote quality

To address the alternative explanation that managements' tax footnote disclosures, rather than the auditors' tax KAMs, drive the valuation of company tax attributes, we re-estimate tests of H1 controlling for tax footnote quality for tax KAM companies and matched sample of nontax-KAM companies. We measure tax footnote quality using the number of sentences (*NUM_SENTENCES*), following Bozanic et al. (2017), Bonsall et al. (2017), and Luo et al. (2023).[33] This subsample consists of 559 tax KAM observations and 401 propensity-score-matched nontax-KAM observations matched without replacement using a 10 percent caliper.[34]

We present results in Table 7. Panel A reports descriptive statistics for the matched sample. Only the values of R&D and CAPEX differ across the treated and matched samples, suggesting that the two groups are similar across an array of company traits. Panel B presents the results of testing H1 with the tax footnote control for both the full matched sample and a balanced sample of 802 tax-KAM and nontax-KAM companies. In both cases, we find that the positive valuation of tax avoidance and deferred tax assets are fully attenuated for tax KAM companies. These findings demonstrate that tax KAMs, rather than management disclosures, drive the results for H1. In an untabulated analysis, we repeat this testing for subtypes of tax KAMs and continue to find that receiving a tax KAM related to uncertain tax positions or deferred tax assets fully attenuates the positive valuation of deferred tax assets. We do not find a significant association between tax avoidance and Tobin's Q for either tax-KAM or nontax-KAM companies in the smaller matched sample.

### 5.3 Textual attributes of tax KAMs

To determine whether textual characteristics of tax KAM disclosures moderate the effect of tax KAMs on the valuation of tax attributes, we group our sample of tax KAM companies based on (1) the degree of negative tone of the tax KAM and (2) the readability of the KAM. Specifically, we create indicator variables equal to one if the tax KAM is above the median in terms of these two textual features. We then re-estimate Model (1), partitioning our full sample of tax KAM observations based on these indicators. Appendix E Table 11 presents the results. For negative tone (readability), Columns 1 and 2 (5 and 6) display results of a model including only industry and year fixed effects, and Columns 3 and 4 (7 and 8) add the full suite of controls. In all specifications, we find that more negative tone attenuates the positive valuation of deferred tax assets ($p < 0.05$) but does not appear to influence the valuation of companies' tax avoidance. With respect to tax KAM

[33] We also measure tax footnote quality using the number of words and FOG index and find that our results endure (untabulated).

[34] Financial statements during our sample period do not include XBRL tags, which requires us to hand-collect tax footnotes. We therefore limit this analysis to this matched subsample of firms.

**Table 7** Controlling for Tax Footnote Disclosure Quality

**Panel A: Descriptive Statistics: Tax KAM versus No Tax KAM – Propensity Score Matched Sample**

| | **With Tax KAM** | | | | **Without Tax KAM** | | | | **Diff. in Means** |
|---|---|---|---|---|---|---|---|---|---|
| **Variable** | **N** | **Mean** | **Median** | **Std. Dev** | **N** | **Mean** | **Median** | **Std. Dev** | **p-value** |
| TOBIN'S_Q | 401 | 1.776 | 1.529 | 0.881 | 401 | 1.744 | 1.365 | 1.061 | 0.643 |
| AVOID | 401 | − 0.249 | − 0.216 | 0.240 | 401 | − 0.263 | − 0.211 | 0.263 | 0.414 |
| DTA_NET | 401 | − 0.007 | − 0.004 | 0.033 | 401 | − 0.007 | − 0.001 | 0.029 | 0.998 |
| NUM_SENTENCES | 401 | 28.436 | 24 | 16.051 | 401 | 27.386 | 22.5 | 17.279 | 0.372 |
| ETR_VOL | 401 | 0.124 | 0.063 | 0.152 | 401 | 0.143 | 0.053 | 0.178 | 0.120 |
| MATERIALITY | 401 | 0.015 | 0.013 | 0.020 | 401 | 0.013 | 0.010 | 0.021 | 0.232 |
| HAVEN | 401 | 0.112 | 0 | 0.316 | 401 | 0.080 | 0 | 0.271 | 0.117 |
| FOREIGN | 401 | 60.617 | 73.259 | 36.546 | 401 | 58.999 | 68.851 | 35.688 | 0.526 |
| PRIORLOSS | 401 | 0.444 | 0 | 0.497 | 401 | 0.423 | 0 | 0.495 | 0.549 |
| NOL | 401 | 0.130 | 0 | 0.336 | 401 | 0.152 | 0 | 0.359 | 0.369 |
| LN_SALES | 401 | 7.180 | 7.156 | 1.534 | 401 | 7.232 | 7.281 | 1.896 | 0.668 |
| SALES_GROWTH | 401 | 0.085 | 0.003 | 0.810 | 401 | 0.130 | 0.035 | 0.699 | 0.400 |
| RETURN_VOL | 401 | 0.086 | 0.077 | 0.037 | 401 | 0.084 | 0.074 | 0.040 | 0.610 |
| LEV | 401 | 0.246 | 0.234 | 0.201 | 401 | 0.239 | 0.212 | 0.216 | 0.645 |
| INTANGIBLE | 401 | 0.265 | 0.238 | 0.245 | 401 | 0.251 | 0.157 | 0.305 | 0.486 |
| FCF | 401 | 0.049 | 0.047 | 0.067 | 401 | 0.045 | 0.042 | 0.066 | 0.379 |
| R&D | 401 | 0.012 | 0 | 0.030 | 401 | 0.008 | 0 | 0.023 | 0.026** |
| CAPEX | 401 | 0.054 | 0.045 | 0.051 | 401 | 0.042 | 0.028 | 0.046 | 0.003*** |
| ACCRUALS | 401 | − 0.049 | − 0.044 | 0.085 | 401 | − 0.043 | − 0.033 | 0.086 | 0.291 |
| VOLUME_VOL | 401 | 8.885 | 8.944 | 1.168 | 401 | 8.751 | 8.628 | 1.640 | 0.183 |

**Table 7** (continued)

**Panel B: Valuation of Company Tax Attributes Controlling for Tax Footnote Quality**

| | **(1)** | **(2)** | **(3)** | **(4)** |
|---|---|---|---|---|
| | **Unbalanced** | | **Balanced** | |
| VARIABLES | No Tax KAM TOBIN'S Q | Tax KAM TOBIN'S Q | No Tax KAM TOBIN'S Q | Tax KAM TOBIN'S Q |
| AVOID | 0.217* | − 0.034 | 0.217* | − 0.000 |
| | (1.906) | (− 0.296) | (1.906) | (− 0.003) |
| DTA_NET | 4.382*** | 0.997 | 4.382*** | 1.587 |
| | (4.085) | (0.746) | (4.085) | (1.104) |
| NUM_SENTENCES | 0.000 | 0.004 | 0.000 | 0.003 |
| | (0.041) | (1.697) | (0.041) | (1.036) |
| ETR_VOL | − 0.187 | − 0.139 | − 0.187 | − 0.004 |
| | (− 1.149) | (− 0.691) | (− 1.149) | (− 0.016) |
| HAVEN | 0.024 | 0.225 | 0.024 | 0.157 |
| | (0.245) | (1.384) | (0.245) | (1.167) |
| FOREIGN | 0.002 | 0.001 | 0.002 | 0.002 |
| | (1.462) | (0.390) | (1.462) | (0.893) |
| PRIORLOSS | − 0.129 | 0.043 | − 0.129 | − 0.022 |
| | (− 1.206) | (0.345) | (− 1.206) | (− 0.150) |
| MATERIALITY | 17.554*** | 11.298** | 17.554*** | 10.901** |
| | (4.545) | (2.961) | (4.545) | (2.332) |
| NOL | 0.252 | 0.213 | 0.252 | 0.227 |
| | (1.519) | (1.645) | (1.519) | (1.666) |
| SALES | − 0.011 | − 0.009 | − 0.011 | − 0.029 |
| | (− 0.448) | (− 0.297) | (− 0.448) | (− 0.693) |
| SALES_GROWTH | 0.033 | − 0.038 | 0.033 | − 0.026 |
| | (1.028) | (− 1.479) | (1.028) | (− 1.540) |
| RETURN_VOL | 0.763 | − 2.914*** | 0.763 | − 2.495** |
| | (0.451) | (− 3.150) | (0.451) | (− 2.481) |

**Table 7** (continued)

| | | | | |
|---|---|---|---|---|
| LEVERAGE | − 0.110 | − 0.038 | − 0.110 | − 0.282 |
| | (− 0.289) | (− 0.200) | (− 0.289) | (− 1.120) |
| INTANGIBLE | 0.220 | 0.079 | 0.220 | 0.170 |
| | (0.751) | (0.403) | (0.751) | (1.052) |
| FCF | 2.834** | 7.094*** | 2.834** | 6.283*** |
| | (2.421) | (7.590) | (2.421) | (3.991) |
| R&D | 8.415 | 0.895 | 8.415 | 1.013 |
| | (1.655) | (0.591) | (1.655) | (0.620) |
| CAPEX | 5.515* | 5.259** | 5.515* | 3.780 |
| | (2.203) | (2.263) | (2.203) | (1.662) |
| ACCRUALS | 0.919* | 1.289** | 0.919* | 0.614 |
| | (1.840) | (2.420) | (1.840) | (0.673) |
| VOLUME_VOL | 0.006 | − 0.066 | 0.006 | − 0.054 |
| | (0.133) | (− 1.308) | (0.133) | (− 1.350) |
| CONSTANT | 0.670 | 1.937*** | 0.670 | 2.235*** |
| | (1.376) | (5.777) | (1.376) | (7.124) |
| Year and Industry FE | YES | YES | YES | YES |
| Observations | 401 | 559 | 401 | 401 |
| R-squared | 0.662 | 0.647 | 0.662 | 0.603 |
| **Tests of Differences Between Models** | | | | |
| Column (1/3) = Column (2/4) | Difference | Chi2 | Difference | Chi2 |
| AVOID | 0.251** | 5.90 | 0.217* | 2.76 |
| DTA_NET | 3.385*** | 6.74 | 2.795* | 3.25 |

Panel A: This table presents descriptive statistics for our analysis of H1(Model (1)) using a balanced propensity score matched sample of tax-KAM and nontax-KAM firms after including a control for tax footnote quality (NUM_SENTENCES). The symbols ***, **, and * represent two-tailed p-values significant at the 0.01, 0.05 and 0.10 levels, respectively

Panel B: This table presents the results of our analysis of H1(Model (1)) using a propensity score matched sample of tax-KAM and nontax-KAM firms after including a control for tax footnote quality (NUM_SENTENCES). We use an OLS regression of company value (TOBIN'S Q) on companies' tax attributes (AVOID and DTA_NET) and a set of control variables for both an unbalanced matched sample (columns 1 and 2) and a balanced matched sample (columns 3 and 4). Tests of differences between coefficients on tax attributes are presented at the bottom of this table. In all regressions, standard errors are clustered by company. Two-tailed t-statistics are presented in brackets, with Chi2 statistics presented in tests of differences in coefficients. The symbols ***, **, and * represent two-tailed p-values significant at the 0.01, 0.05 and 0.10 levels, respectively readability, we find that more readable tax KAMs decrease the valuation of tax avoidance (Columns 5 and 6 only), potentially suggesting a benefit to the client from obfuscation of tax issues by the auditor.

In Table 12, we present associations between tax KAM readability and negative tone and current and future purchases of APTS. We find that more readable tax KAMs are associated with lower concurrent and future APTS, consistent with the auditor being rewarded for using more obscure and confusing language (Sulcaj 2023). Further, we find that more negatively toned tax KAMs are associated with decreases in concurrent and future APTS (Columns 3 and 6), suggesting auditors are punished for issuing more pessimistically worded tax KAMs.

### 5.4 Changes in tax avoidance

Another potential way to resolve tax KAMs is to reduce tax avoidance to decrease the estimation uncertainty that causes the tax KAM. We explore this possibility by replacing the dependent variable in Model (4) with one-year changes in both the cash and GAAP ETR. We find no significant associations between changes in tax avoidance and tax KAM resolution (untabulated). This suggests either managers do not believe that the costs of tax KAM disclosure outweigh the benefits of tax avoidance or that it is infeasible to change the complex tax planning structures that can drive tax complexity and avoidance (Balakrishnan et al. 2019).

## 6 Robustness checks

### 6.1 Loss companies

We examine whether our results differ among companies that do not generate taxable income. These companies are less likely to prioritize tax avoidance, as they are unlikely to derive cash tax savings from doing so. Further, because tax planning among loss companies is less likely to materially affect financial statement income, we expect that auditors will assess less risk of material misstatement in these companies' tax accounts. We find that the market positively values the tax avoidance of prior loss companies without tax KAMs but that this valuation is fully attenuated for tax KAM companies. We find that the market does not positively value the deferred tax assets of prior loss companies with or without tax KAMs, consistent with the likelihood that they will not generate sufficient taxable income to realize the deferred tax assets. Inferences for the consequences of tax KAMs for APTS are unchanged in the prior loss sample (untabulated).

### 6.2 Other robustness tests

We perform several additional robustness tests. First, we conduct a series of falsification tests in which we replace our tax KAM variables with indicators for other KAM topics (i.e., goodwill, revenue recognition, mergers and acquisitions, and internal controls) to test our conjecture that the consequences of KAMs vary by topic. We find that these KAMs do not affect the valuation of company tax attributes, resolving these KAMs is not associated with increased APTS and companies' tax attributes do not predict these KAM types.[35,36] In a further falsification test, we replicate our APTS analyses using changes in audit fees, total non-audit service fees, and nontax non-audit service fees instead of APTS. We find no associations significant at the $p \leq 0.10$ level between *RESOLVE_TAXKAM* and any of these three alternative fee measures (untabulated). We find no significant associations between *EXPECT_TAXKAM* and both non-audit service fee measures but a positive and significant association with audit fees (untabulated). Finally, all results are also robust to controlling for goodwill impairments and impairment KAMs (untabulated).[37]

To rule out the possibility that functional form misspecification leads to erroneous inferences in our analysis of the relation between resolving a tax KAM and APTS, we employ an entropy-balanced model (Hainmueller 2012; Shipman et al. 2017). We match each tax-KAM company-year to a nontax-KAM control company-year on the independent variables from our expectations model based on the first, second, and third moments. We replicate our resolve tax-KAM tests using these weights. For our full sample, our findings for *%ΔAPTS* are fully robust (untabulated). For the sample of companies that receive a tax KAM at least once during the sample period, the results are robust to both *%ΔAPTS* and *ΔLN_APTS* (untabulated).

We acknowledge that companies headquartered in tax havens may have different tax KAM incentives than other companies related to the valuation of their tax attributes and reputation for tax avoidance. We therefore re-estimate Models (1), (2), and (4) after dropping companies headquartered in tax havens. Our inferences are unchanged (untabulated). Given the literature demonstrating that individual audit partners have unique styles for KAM reporting (Rousseau and Zehms 2024) and fee premia (Taylor 2011), we also repeat our tests of H2 after controlling for changes in audit partner. Our inferences endure (untabulated). Finally, we test the alternative explanation that clients could resolve tax KAMs by reducing their tax avoidance by re-estimating Models (2) and (4) with changes in GAAP and cash effective tax rates as dependent variables. We do not find any significant associations (untabulated).

[35] The valuation of deferred tax assets is lower for firms receiving a KAM related to mergers and acquisitions, but this difference is statistically significant only in one-tailed testing ($p < 0.10$).

[36] We also document significantly lower area under receiver operating characteristic curve for these falsification tests, ranging from 0.74 to 0.86, suggesting decreased explanatory power for the issuance of nontax KAMs.

[37] Given extreme values of ETR are associated with valuation allowances and goodwill impairments (Schwab et al. 2022), we also confirm our primary results are robust to excluding observations with ETRs equal to 0 and 1 (untabulated).

## 7 Conclusion

This study examines a prevalent and economically important type of KAM to provide insights about expanded audit reporting. Our results suggest that tax KAMs fully attenuate the positive valuation of companies' tax avoidance and deferred tax assets. This valuation consequence incentivizes managers to avoid receiving tax KAMs, increasing their incentives to purchase APTS. Specifically, auditors are more likely to stop issuing tax KAMs for companies that subsequently increase their fees for APTS, suggesting a detrimental effect of economic bond incentives on the objectivity of the auditor's tax KAM decision.

Our study makes numerous contributions to research and practice. First, we reveal that a potential consequence of KAMs is to increase clients' demand for non-audit services in a manner that can raise questions about the independence of auditors' KAM judgments. This finding enhances the profession's understanding of two timely regulatory priorities: expanded audit reporting and the resurgence of auditors' advisory practices. Second, we provide insight into the conflicted literature on the consequences of expanded audit reporting. We illustrate that the answers to these questions will likely vary with the subject matter of the KAMs, including the specific issues cited within a single KAM type, and demonstrate the need to use variables that tightly map to the topic of the KAM.

Our inferences are subject to several caveats. First, we recognize that KAM information is likely not incremental to other information the company has already made public.[38] Instead, we believe tax KAMs provide a confirmatory signal about the riskiness of existing tax disclosures and activities from an auditor with high source credibility. Second, we cannot rule out self-selection issues related to KAM disclosures. We alleviate these concerns by illustrating the robustness of our results to a subsample of only those companies that receive tax KAMs and our entropy balancing tests, but we cannot rule out this issue. Third, we acknowledge that our findings concerning U.K. KAM reporting may not generalize to the United States due to differences in KAM and CAM standards and the broader regulatory environment (Lohwasser et al. 2024). However, we expect that our findings will generalize to IAASB jurisdictions such as Europe, Asia, and Australasia because the United Kingdom expanded reporting requirements are almost identical to IAASB KAMs and these regions have more comparable litigation environments. Fourth, given our sample consists of large London Stock Exchange premium segment companies that mostly employ Big Four auditors, we cannot speak

[38] The quantity and varied forums of companies' public disclosures mean that it would be extremely costly to attempt to hand-collect data to test whether tax KAMs contain incremental information and that attempting to do so is unlikely to yield sufficient controls to permit valid causal inference. Further, the contribution associated with such efforts is ambiguous, given the many existing studies examining stock market reactions to KAM adoption (Bédard et al. 2019; Gutierrez et al. 2018; Liao et al. 2019; Burke et al. 2023; Lennox et al. 2023; Goh et al. 2023).

to the effects in smaller companies, which may have poorer information environments. Finally, our findings illustrate that one prominent type of KAM has consequences, but we cannot state that this is true for all types of KAMs. Our study motivates further research into the consequences of other KAM topics and suggests opportunities for future research on the implications of KAM reporting for demand for auditor-provided tax services.

## Appendix A

To identify situations in which companies have a high probability of receiving a tax KAM but do not have one, we build an expectations model for tax KAM disclosure using the following probit regression, which includes industry ($\mu_{ind}$) and year ($\tau_t$) fixed effects:

$$\begin{aligned} TaxKAM_{i,t} = {} & \gamma_0 + \gamma_1 AVOID_{,t} + \gamma_2 ETR\_VOL_{i,t} + \gamma_3 DTA\_NET_{,t} + \gamma_4 FOREIGN_{i,t} \\ & + \gamma_5 HAVEN_{i,t} + \gamma_6 MATERIALITY_{i,t} + +\gamma_7 BIG\_4_{i,t} + \gamma_8 NOL_{,t} + \gamma_9 PRIORLOSS_{i,t} \\ & + \gamma_{10} LN\_AUDIT\_FEES_{i,t} + \gamma_{11} LN\_APTS_{i,t} + \gamma_{12} LN\_NONTAX\_NAS_{i,t} \\ & + \gamma_{13} MERGE\_RESTRUCT_{i,t} + \gamma_{14} ASSETS_{i,t} + \gamma_{15} PTI_{i,t} + \gamma_{16} FCF_{i,t} \\ & + \gamma_{17} NUM\_KAMS_{i,t} + \gamma_{18} KAM\_REGPERIOD_{i,t} + \mu_{ind} + \tau_t + \varepsilon_{i,t} \end{aligned} \tag{5}$$

*TaxKAM* is an indicator variable equal to one if the company receives a tax KAM in year t and zero otherwise. Research suggests that the complexity associated with tax planning, such as income-shifting to low-tax jurisdictions, increases organizational and financial reporting complexity (Lassila et al. 2010; Blouin and Krull 2018; Chen et al. 2018; Balakrishnan et al. 2019; Frank et al. 2009). Further, there is significant subjectivity and uncertainty in the realizability of deferred tax assets, future deductible amounts that are not realized unless the company generates future taxable income (Amir et al. 1997; 2001; Guenther and Sansing 2000; Flagmeier 2022). *AVOID*, *ETR_VOL*, *DTA_NET*, *FOREIGN*, and *HAVEN* capture the tax characteristics of the company and are defined in Model (1).

We also control for auditor involvement with the tax function using the natural log of auditor-provided tax services (*LN_APTS*). If auditor-provided tax services are negatively associated with tax KAM disclosure, this could imply a threat to auditor independence, where the auditor does not issue a tax KAM to preserve these additional fees. However, it could also indicate a knowledge spillover effect, where the auditor is more comfortable with the tax function and determines it does not meet the criteria for KAM disclosure (i.e., it is not one of the most challenging issues the auditor faced in the audit).

We control for auditor effort using audit fees (*LN_AUDIT_FEES*). We control for mergers and restructurings with the indicator *MERGE_RESTRUCT*, which is equal to one if the company reports expenditures related to mergers, acquisitions, or restructuring. We also include controls for the provision of nontax-related non-audit services (*LN_NONTAX_NAS*), net operating losses (*NOL* and *PRIORLOSS*), materiality of tax expense (*MATERIALITY*), company size (*ASSETS*), profitability (*PTI*), free cash flow (*FCF*), auditor size (*BIG_4*), and the total number of KAMs in the company's audit report (*NUM_KAMS*). Finally, we control for the alignment of risk of material misstatement disclosures with KAM reporting under IAASB regulations after June 15, 2017 (*KAM_REGPERIOD*).

Our sample includes 4,008 observations from 766 unique companies with available data. Table 8 describes the construction of our sample. Table 9 Panel A presents descriptive statistics for the full sample. Table 9 Panel B presents descriptive statistics comparing companies with and without tax KAMs.

Table 10 presents the results of estimating Model (5). The expectations model performs well in predicting tax KAMs, with the area under curve of roughly 0.90, suggesting excellent discriminatory power (Hosmer and Lemeshow 2000). The discriminatory power of our tests also remains high when excluding industry and year fixed effects, with area under the receiver operating characteristic curve ranging from 0.889 to 0.891. We use the predicted value of *TAX_KAM* from this expectations model (*EXPECT_TAXKAM)* in our hypothesis testing.

**Table 8** Sample Selection

| **Panel A: Expectations Model** | | |
|---|---|---|
| **Total firm/year KAM observations** | | 7,339 |
| Less: | Observations without Factset identifiers | (1,012) |
| | Missing currency conversion data | (61) |
| | Missing book ETR data | (11) |
| | Missing audit and non-audit fees data | (42) |
| | Missing balance sheet controls | (51) |
| | Missing cash flow statement controls | (134) |
| | Non-Main Market, premium segment firms | (2,020) |
| Sample for expectations model | | 4,008 |
| Number of unique companies included in expectations model | | 766 |

This table reports the construction of our sample for the expectations model of tax KAM issuance (model (5))

**Table 9** Descriptive Statistics for Expectations Model

**Panel A: Expectations Model Descriptives**

| Variable | N | Mean | Std. Dev | P25 | Median | P75 |
|---|---|---|---|---|---|---|
| TAX_KAM | 4008 | 0.169 | 0.375 | 0 | 0 | 0 |
| AVOID | 4008 | - 0.209 | 0.288 | - 0.247 | - 0.143 | - 0.000 |
| AVOID_3YR | 4008 | - 0.271 | 0.349 | - 0.286 | - 0.174 | - 0.002 |
| ETR_VOL | 4008 | 0.140 | 0.202 | 0.005 | 0.032 | 0.171 |
| DTA_NET | 4008 | - 0.005 | 0.027 | - 0.008 | 0 | 0 |
| HAVEN | 4008 | 0.115 | 0.319 | 0 | 0 | 0 |
| TAX_MATERIALITY | 4008 | 0.010 | 0.019 | 0.000 | 0.003 | 0.016 |
| FOREIGN | 4008 | 30.359 | 38.776 | 0 | 0.339 | 68.077 |
| BIG 4 | 4008 | 0.841 | 0.366 | 1 | 1 | 1 |
| NOL | 4008 | 0.108 | 0.310 | 0 | 0 | 0 |
| PRIORLOSS | 4008 | 0.569 | 0.495 | 0 | 1 | 1 |
| LN_AUDIT FEES | 4008 | 12.552 | 1.783 | 10.700 | 12.568 | 13.804 |
| LN_APTS | 4008 | 3.795 | 5.357 | 0 | 0 | 9.736 |
| LN_NONTAXNAS | 4008 | 8.054 | 5.656 | 0 | 10.591 | 12.472 |
| MERGE_RESTRUCT | 4008 | 0.428 | 0.495 | 0 | 0 | 1 |
| ASSETS | 4008 | 6.662 | 1.973 | 5.447 | 6.616 | 7.824 |
| PTI | 4008 | 0.064 | 0.175 | 0.015 | 0.062 | 0.124 |
| FCF | 4008 | 0.025 | 0.116 | 0 | 0.021 | 0.057 |
| NUMBER OF KAMS | 4008 | 2.980 | 1.476 | 2 | 3 | 4 |
| KAM_REGPERIOD | 4008 | 0.481 | 0.500 | 0 | 0 | 1 |

**Panel B: Expectations Model: Companies With and Without Tax KAMs**

| | With Tax KAMs | | | Without Tax KAMs | | | |
|---|---|---|---|---|---|---|---|
| **Variable** | **N** | **Mean** | **Std. Dev** | **N** | **Mean** | **Std. Dev** | **p-value** |
| AVOID | 677 | - 0.262 | 0.259 | 3331 | - 0.198 | 0.292 | 0.000*** |
| AVOID_3YR | 677 | - 0.313 | 0.318 | 3331 | - 0.263 | 0.354 | 0.001*** |
| ETR_VOL | 677 | 0.147 | 0.170 | 3331 | 0.139 | 0.208 | 0.346 |
| DTA_NET | 677 | - 0.006 | 0.036 | 3331 | - 0.005 | 0.025 | 0.148 |
| HAVEN | 677 | 0.095 | 0.293 | 3331 | 0.119 | 0.324 | 0.067* |
| TAX_MATERIALITY | 677 | 0.014 | 0.021 | 3331 | 0.009 | 0.018 | 0.000*** |
| FOREIGN | 677 | 64.793 | 35.178 | 3331 | 23.361 | 35.609 | 0.000*** |
| BIG 4 | 677 | 0.981 | 0.137 | 3331 | 0.813 | 0.390 | 0.000*** |
| NOL | 677 | 0.146 | 0.354 | 3331 | 0.100 | 0.300 | 0.000*** |
| PRIORLOSS | 677 | 0.471 | 0.500 | 3331 | 0.588 | 0.492 | 0.000*** |
| LN_AUDIT FEES | 677 | 14.439 | 1.288 | 3331 | 12.168 | 1.618 | 0.000*** |
| LN_APTS | 677 | 6.179 | 6.241 | 3331 | 3.310 | 5.023 | 0.000*** |
| LN_NONTAXNAS | 677 | 11.690 | 4.088 | 3331 | 7.316 | 5.645 | 0.000*** |
| MERGE_RESTRUCT | 677 | 0.770 | 0.421 | 3331 | 0.358 | 0.480 | 0.000*** |
| ASSETS | 677 | 8.124 | 1.699 | 3331 | 6.365 | 1.891 | 0.000*** |

**Table 9** (continued)

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| PTI | 677 | 0.066 | 0.099 | 3331 | 0.064 | 0.187 | 0.796 |
| FCF | 677 | 0.047 | 0.064 | 3331 | 0.021 | 0.124 | 0.000*** |
| NUMBER OF KAMS | 677 | 4.288 | 1.430 | 3331 | 2.713 | 1.337 | 0.000*** |
| KAM_REGPERIOD | 677 | 0.380 | 0.486 | 3331 | 0.502 | 0.500 | 0.000*** |

This table presents descriptive statistics for our tests of Model (5). Panel A describes the full sample. Panel B presents comparisons of tax KAM companies and nontax KAM companies. The symbols ***, **, and * represent two-tailed p-values significant at the 0.01, 0.05 and 0.10 levels, respectively

**Table 10** Predicting Tax KAMs

| VARIABLES | (1)<br>TAX_KAM | (2)<br>TAX_KAM |
|---|---|---|
| AVOID | 0.053* | |
| | (1.787) | |
| 3YR _AVOID | | 0.068*** |
| | | (2.986) |
| ETR_VOL | 0.000 | 0.029 |
| | (0.004) | (0.722) |
| DTA_NET | 0.563** | 0.575** |
| | (2.305) | (2.370) |
| FOREIGN | 0.001*** | 0.001*** |
| | (5.098) | (5.194) |
| HAVEN | - 0.005 | - 0.005 |
| | (- 0.160) | (- 0.154) |
| MATERIALITY | - 0.193 | - 0.136 |
| | (- 0.403) | (- 0.294) |
| BIG_4 | 0.090*** | 0.089*** |
| | (2.938) | (2.944) |
| NOL | - 0.013 | - 0.002 |
| | (- 0.594) | (- 0.127) |
| PRIORLOSS | 0.019 | 0.026 |
| | (1.007) | (1.357) |
| LN_AUDIT FEES | 0.041*** | 0.041*** |
| | (3.341) | (3.373) |
| LN_APTS | 0.001 | 0.001 |
| | (0.393) | (0.435) |
| LN_NONTAX_NAS | 0.000 | 0.000 |
| | (0.058) | (0.051) |
| MERGE_RESTRUCT | - 0.011 | - 0.009 |
| | (- 0.719) | (- 0.628) |
| ASSETS | 0.007 | 0.006 |
| | (0.732) | (0.704) |
| PTI | 0.027 | 0.041 |
| | (0.338) | (0.540) |

**Table 10** (continued)

| VARIABLES | (1)<br>TAX_KAM | (2)<br>TAX_KAM |
|---|---|---|
| FCF | 0.104 | 0.101 |
| | (0.957) | (0.940) |
| NUM_KAMS | 0.036*** | 0.036*** |
| | (5.969) | (5.967) |
| KAM_REGPERIOD | 0.016 | 0.017 |
| | (0.631) | (0.648) |
| Year FE | YES | YES |
| Industry FE | YES | YES |
| Observations | 4,008 | 4,008 |
| Area Under Curve | 0.898 | 0.899 |

This table presents the results of estimating Model (5), which predicts the likelihood that a company will receive a tax KAM using probit regression with standard errors clustered by firm. Coefficients represent average marginal effects. Two-tailed t-statistics are presented in brackets. The symbols ***, **, and * represent two-tailed p-values significant at the 0.01, 0.05 and 0.10 levels, respectively

## Appendix B

### Example of a Tax KAM from KPMG's 2019 Audit Report for Unilever PLC

Title: Uncertain direct tax transfer pricing provisions.

Description:
The Unilever Group has extensive international operations and is operating in a number of tax jurisdictions, each with its own taxation regime. The laws and regulations for transfer pricing in each jurisdiction are open to different interpretations by taxpayers and tax authorities and require judgement in the interpretation thereof. Judgements are made by the Unilever Group in assessing the potential outcome of investigations by the authorities, and if a liability exists. We identified the assessment of uncertain direct tax transfer pricing provisions as a key audit matter. Due to the complex nature of transfer pricing across multiple jurisdictions, there is judgement applied by the Unilever Group with respect to interpretations of the tax legislation and to assess the potential outcome of investigations by the authorities. Complex auditor's judgement was also required in assessing the potential outcome of investigations by the authorities.

Response:
The primary procedures performed to address this key audit matter included the following:

- Tested certain controls within the income tax process including controls around the assessment of the potential outcome of investigations, the completeness of the exposures and the recording and re-assessments of transfer pricing provisions.

- We involved tax professionals with specialized skills and knowledge to assist in assessing changes to the transfer pricing models for compliance with applicable laws and regulations; and evaluating a sample of exposures using our own expectations based on our knowledge of the Unilever Group, considering relevant judgements passed and investigations by authorities, related correspondence with the tax authorities as well as inspecting relevant tax opinions from third parties.
- Assessed the Group's disclosures in respect of direct tax provisions and uncertain tax positions.

Conclusion:
The results of our testing were satisfactory (2018: satisfactory) and we found the level of direct tax provisions to be acceptable (2018: acceptable).

## Appendix C

### Variable Definitions

| Variable | Definition |
|---|---|
| **Valuation Model** | |
| *TaxKAM* | Indicator variable equal to one if the firm received a tax KAM in year t, zero otherwise |
| *YHAT > 0.75* | Indicator variable equal to one if the predicted value of tax KAM issuance (*EXPECT_TAXKAM*) exceeds 0.75, zero otherwise |
| *TOBIN'S Q* | Market value of assets divided by book value of assets [ff_mkt_val + ff_assets – (ff_bps*ff_com_shs_out)]/ff_assets |
| *AVOID* | Income tax expense (ff_inc_tax) divided by pretax book income (ff_ptx_inc), multiplied by negative 1 |
| *DTA_NET* | Total deferred tax assets (ff_dfd_tax_db) less total deferred tax liabilities (ff_dfd_tax_cr) scaled by total assets (ff_assets) |
| *ETR_VOL* | Standard deviation of *ETR* over the period t- 2 through t |
| *FOREIGN* | Percentage of sales derived from foreign sources (ff_for_sales_pct) |
| *HAVEN* | Indicator variable equal to one if the issuer is headquartered in a tax haven country, as designated by Dyreng and Lindsey (2009), zero otherwise |
| *PRIORLOSS* | Indicator variable equal to one if the firm has negative pretax income (ff_ptx_inc) during the current or prior five years, zero otherwise |
| *MATERIALITY* | Income tax expense (ff_inc_tax) scaled by total assets (ff_assets) |

| Variable | Definition |
|---|---|
| *NOL* | Indicator variable equal to one if the firm has negative income tax expense (ff_inc_tax), zero otherwise |
| *SALES* | Natural log of one plus total sales (ff_sales) |
| *SALES_GROWTH* | Average sales (ff_sales) growth over the past three years |
| *RETURN_VOL* | Standard deviation of monthly stock returns over prior 36 months |
| *LEVERAGE* | Total debt (ff_debt_lt + ff_debt_st_tot) scaled by total assets (ff_assets) |
| *INTANGIBLE* | Total intangible assets (ff_intang) scaled by total assets (ff_assets) |
| *FCF* | Operating cash flow (ff_oper_cf) less capital expenditures (ff_capex), scaled by total assets (ff_assets) |
| *R&D* | Research and development costs (ff_rd_exp) scaled by total assets (ff_assets) |
| *CAPEX* | Capital expenditures scaled by total assets (ff_assets) |
| *ACCRUALS* | Net income before extraordinary items (ff_net_inc – ff_xord) less operating cash flow (ff_oper_cf), scaled by total assets (ff_assets) |
| *VOLUME_VOL* | Log of the standard deviation of monthly trading volume over prior 36 months |
| **Auditor-Provided Tax Services Models** (repeated variables defined above) | |
| *ΔLN_APTS* | Change in the log of auditor-provided tax services (APTS), equal to log($APTS_t$ – $APTS_{t-1}$) for concurrent change analysis and log($APTS_{t+1}$ – $APTS_t$) for future change analyses |
| *%ΔAPTS* | Percentage change in APTS fees equal to (($APTS_t$ – $APTS_{t-1}$)/$APTS_{t-1}$) for concurrent change analyses and (($APTS_{t+1}$ – $APTS_t$)/($APTS_t$) for future change analyses |
| *TAXKAM_FIRM* | Indicator variable equal to one if the client receives a tax KAM in year t, year t- 1, or both, zero otherwise |
| *RESOLVE_TAXKAM* | Indicator variable equal to one if the client receives a tax KAM in year t- 1 but does not receive a tax KAM in year t, zero otherwise |
| *EXPECT_TAXKAM* | Predicted value of *TaxKAM* from expectations Model (5) |
| *BIG_4* | Indicator variable equal to one if the firm is audited by a Big Four auditor in year t, zero otherwise |
| *NOL* | Indicator variable equal to one if the firm has negative income tax expense (ff_inc_tax), zero otherwise |
| *MERGE_RESTRUCT* | Indicator variable equal to one if the company has merger, acquisition, or restructuring expenses during the year, zero otherwise |
| *ASSETS* | Log of total assets (ff_assets) |
| *PTI* | Pretax income (ff_ptx_inc) scaled by total assets (ff_assets) |
| *ΔAUDITOR* | Indicator variable equal to one if the audit firm changed, zero otherwise |

| Variable | Definition |
|---|---|
| *BUSY* | Indicator variable equal to one if the firm's year end falls between December and March |
| *INVREC* | Inventory (ff_inven) plus receivables (ff_receiv_tot) scaled by total assets (ff_assets) |
| *MB* | Market value of outstanding shares (ff_price_close_fp*ff_com_shs_out) scaled by book value of common equity (ff_com_eq) |
| *LITIGATE* | Indicator variable equal to one if the firm's primary operations are in a high litigation industry (biotech, computers, electronics, and retail) following Francis et al. (1994) |
| *TAX_EXPERT* | Indicator variable equal to one if the audit office's tax service market share in a given city and two-digit industry SIC exceeds or equals 30 percent, following McGuire et al. (2012). Market share is defined as total tax fees paid to the audit firm divided by total tax fees paid to all other audit firms in the same industry and city |
| *NUM_KAMS* | Total number of KAMs the firm receives in year t |
| *KAM_REGPERIOD* | Indicator variable equal to one if the observation has a year-end after June 15, 2017, the effective date of audit regulation conforming existing risk of material misstatement disclosures more closely to IAASB KAMs |
| **Additional Variables for Supplemental Tests** (repeated variables defined above) | |
| *UTP_KAM* | Indicator variable equal to one if the firm received a tax KAM related to uncertain tax positions in year t, zero otherwise |
| *DTA_KAM* | Indicator variable equal to one if the firm received a tax KAM related to deferred tax assets in year t, zero otherwise |
| *COMP_KAM* | Indicator variable equal to one if the firm received a tax KAM related to multi-jurisdictional complexity in year t, zero otherwise |
| *OTHER_KAM* | Indicator variable equal to one if the firm received a tax KAM related to something other than uncertain tax positions, deferred tax assets, or multi-jurisdictional complexity in year t, zero otherwise |
| *EXPECT_UTPKAM* | Predicted value of *UTP_KAM* from expectations Model (5) |
| *EXPECT_DTAKAM* | Predicted value of *DTA_KAM* from expectations Model (5) |
| *EXPECT_COMPKAM* | Predicted value of *COMP_KAM* from expectations Model (5) |
| *EXPECT_OTHERKAM* | Predicted value of *OTHER_KAM* from expectations Model (5) |
| *READABILITY* | Gunning fog index measuring the years of education required to understand a written text times negative one, such that higher values reflect more readable KAMs |
| *NEG_TONE* | Number of negative words in the tax KAM scaled by total number of words. We measure negative words following the 2018 update to the Loughran and McDonald (2011) word lists |
| *PURCH_APTS* | Indicator variable equal to one if the client purchases APTS in the current year, zero otherwise |

| Variable | Definition |
|---|---|
| **Expectations Model** (repeated variables defined above) | |
| *3YR_AVOID* | Income tax expense (ff_inc_tax) over period t- 2 through t, scaled by pretax book income (ff_ptx_inc) over period t- 2 through t, multiplied by negative 1 |
| *LN_AUDIT_FEES* | Natural log of audit fees |
| *LN_APTS* | Natural log of one plus tax-related service fees |
| *LN_NONTAX_NAS* | Natural log of one plus nontax-related, non-audit service fees |

This appendix presents definitions for all variables used in our analysis. We winsorize all continuous variables at 1% and 99%

## Appendix D

### Hand-Coding of Tax KAMs

To inform our investigation of the determinants of specific types of tax KAMs, we obtained the text of all tax KAMs in our sample from Audit Analytics Datafeeds. Upon manually reviewing a subset of these tax KAMs, we judged that, while the Audit Analytics KAMs database has a high accuracy for distinguishing tax KAMs from other types of KAMs, the classification of those tax KAMs into Audit Analytics' uncertain tax position, deferred taxes, and other tax KAMs categories contained errors and inconsistencies. Specifically, we saw that virtually identical tax KAMs issued by the same auditor for the same client over several years were classified differently, depending on the year. Further, in many cases, the Audit Analytics classification of the tax KAMs differed from the classification the tax scholars on our author team deemed appropriate for that KAM. As a result, we decided to review each in-sample tax KAM and manually classify its subcategory.

To begin, we held a brainstorming session to identify possible subcategories of tax KAMs based on our reading of the tax KAMs so far. We arrived at the following classifications: materiality, multijurisdictional complexity/transfer pricing, specific issues (transactions, investigations, penalties, and uncertain tax positions), Tax Cuts and Jobs Act, deferred tax assets, deferred taxes other than deferred tax assets, and earnings management through the tax account. Next each author reviewed the same 60 tax KAMs and coded them into these categories to ensure high inter-rater reliability and refine our classification scheme based on the trends we identified in our systematic review. After consulting on this hand coding, we agreed on a final categorization scheme comprising uncertain tax positions, multijurisdictional complexity, deferred tax assets, and other tax KAMs.

Using this scheme, we each coded the same 150 tax KAMs, examined and resolved any differences in our categorization, and developed a list of common words and phrases associated with each category to promote consistency in our coding, as below:

| Uncertain Tax Positions | Complexity | Deferred Tax Assets | Other |
|---|---|---|---|
| - Transactions<br>- Investigations<br>- Penalties<br>- Contingencies<br>- Uncertain tax position<br>- Uncertainty with respect to a transaction<br>- Accruals for tax contingencies<br>- Tax payable amount uncertain<br>- Exposures<br>- Compliance | - Generic uncertainty<br>- Multiple jurisdictions<br>- Transfer pricing<br>- Uncertainty in determining the tax provision | - Net operating loss<br>- Deferred tax assets<br>- Valuation allowance | -Any tax KAM not belonging to the other three categories<br>-Any tax KAM referencing a special issue, such as first-time adoption of new regulation |

Finally, two authors divided and coded the remaining in-sample tax KAMs using the above list as a guide and consulting with each other as needed to resolve ambiguities and promote consistency.

Correlations between our hand-coded categorization and the Audit Analytics' categorization is as follows:

| VARIABLES | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| (1) UTP_KAM | 1 | | | | | |
| (2) DTA_KAM | - 0.447* | 1 | | | | |
| (3) COMP_KAM | 0.283* | - 0.213* | 1 | | | |
| (4) OTHER_KAM | - 0.133* | - 0.088* | - 0.072 | 1 | | |
| (5) AuditAnalytics_DeferredTaxes | - 0.496* | 0.600* | - 0.280* | 0.006 | 1 | |
| (6) AuditAnalytics_UTPs | 0.325* | - 0.336* | 0.167* | - 0.013 | - 0.300* | 1 |
| (7) AuditAnalytics_OtherIncomeTaxes | 0.157* | - 0.128* | 0.102* | 0.009 | - 0.496* | - 0.587* |

This table illustrates that our hand-coding yielded significantly different categorizations than Audit Analytics. The highest correlation in the two coding schemes is the 60 percent overlap in our *DTA_KAMs* and *AuditAnalytics_DeferredTaxes.* The *UTP_KAM* and *AuditAnalytics_UTPs* categories are only 33 percent correlated, and the *OTHER_KAM* and *AuditAnalytics_OtherIncomeTaxes* categories are not significantly correlated.

# Appendix E

## Tax KAM Textual Properties

This appendix presents the results supplemental tests of the consequences of tax KAM textual properties on the valuation of company tax attributes and purchases of auditor-provided tax services. We describe these tests and discuss the results in Sect. 5.3 of the manuscript.

**Table 11** Tax KAM Textual Properties and Valuation of Company Tax Attributes

**Panel A: Regression Results**

| | **NEGATIVE TONE** | | | | **READABILITY** | | | |
|---|---|---|---|---|---|---|---|---|
| **VARIABLES** | **(1) NEG_TONE = 0 TOBIN'S Q** | **(2) NEG_TONE = 1 TOBIN'S Q** | **(3) NEG_TONE = 0 TOBIN'S Q** | **(4) NEG_TONE = 1 TOBIN'S Q** | **(5) HIGH_READ = 0 TOBIN'S Q** | **(6) HIGH_READ = 1 TOBIN'S Q** | **(7) HIGH_READ = 0 TOBIN'S Q** | **(8) HIGH_READ = 1 TOBIN'S Q** |
| AVOID | 0.499*** | 0.261** | 0.069 | 0.162 | 0.514*** | 0.060 | 0.062 | - 0.002 |
| | (3.895) | (2.109) | (0.392) | (0.976) | (3.143) | (0.397) | (0.342) | (- 0.009) |
| NET_DTA/DTL | 6.314*** | 0.768 | 4.474*** | 1.246 | 3.979** | 2.550 | 2.131 | 3.822*** |
| | (4.787) | (0.705) | (4.216) | (1.215) | (2.334) | (1.501) | (1.629) | (2.975) |
| Controls | NO | NO | YES | YES | YES | NO | YES | YES |
| Year and Industry FE | YES | YES | YES | YES | YES | YES | YES | YES |
| Observations | 332 | 331 | 317 | 316 | 332 | 331 | 317 | 316 |
| R-squared | 0.209 | 0.225 | 0.679 | 0.544 | 0.269 | 0.199 | 0.611 | 0.627 |

**Table 11** (continued)

| Panel B: Tests of Differences Between Models | Difference | p-value |
|---|---|---|
| **NEGATIVE TONE:** Column (1) = Column (2) | | |
| AVOID | 0.238 | (0.168) |
| DTA_NET | 5.546*** | (0.000) |
| **NEGATIVE TONE:** Column (3) = Column (4) | | |
| AVOID | - 0.093 | (0.312) |
| DTA_NET | 3.228** | (0.020) |
| **READABILITY**: Column (5) = Column (6) | | |
| AVOID | 0.474** | (0.027) |
| DTA_NET | 1.429 | (0.432) |
| **READABILITY:** Column (2) = Column (4) | | |
| AVOID | - 0.064 | (0.787) |
| DTA_NET | - 1.691 | (0.300) |

This table reports differences in the valuation of tax attributes among tax KAM firms across different textual characteristics of the tax KAMs. Columns 1–4 present results for KAMs with and without a negative tone, and columns 5–8 present results for KAMs with differing levels of readability. Panel B presents tests of differences in coefficients across the various subsamples. Untabulated control variables include ETR_VOL, HAVEN, FOREIGN, PRIORLOSS, MATERIALITY, NOL, SALES, SALES_GROWTH, RETURN_VOL, LEVERAGE, INTANGIBLE, FCF, R&D, CAPEX, ACCRUAL, and VOLUME_VOL. We use an OLS regression with standard errors clustered by firm. Two-tailed t-statistics are presented in brackets. The symbols ***, **, and * represent two-tailed p-values significant at the 0.01, 0.05 and 0.10 levels, respectively

**Table 12** Tax KAM Textual Properties and Auditor-Provided Tax Services

| | ***Concurrent* (t)** | | | **Future (t + 1)** | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| VARIABLES | LN_APTS | PURCH_ APTS | ΔLN_ $APTS_{i,t-1,t}$ | LN_APTS | PURCH_ APTS | ΔLN_$APTS_{i,t,t+1}$ |
| READABIL- ITY | - 0.342*** | - 0.030*** | - 0.024 | - 0.288* | - 0.026** | - 0.024 |
| | (- 3.08) | (- 3.32) | (- 0.32) | (- 1.96) | (- 2.19) | (- 0.28) |
| Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Year and Industry FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 501 | 501 | 501 | 350 | 350 | 350 |
| R-squared | 0.417 | 0.374 | 0.137 | 0.405 | 0.368 | 0.163 |
| NEG_TONE | - 2.579 | - 0.012 | - 45.613*** | - 20.508 | - 1.597 | - 45.137** |
| | (- 0.09) | (- 0.01) | (- 2.86) | (- 0.60) | (- 0.57) | (- 2.07) |
| Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Year and Industry FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 501 | 501 | 501 | 350 | 350 | 350 |
| R-squared | 0.404 | 0.358 | 0.144 | 0.397 | 0.357 | 0.169 |

This table reports the association between the readability and tone of tax KAMs and client companies' purchases of auditor-provided tax services. Columns 1–3 present associations between tax KAM properties and concurrent purchases of auditor-provided tax services, while Columns 4–6 present associations with next-period purchases of auditor-provided tax services. We use ordinary least squares regression with standard errors clustered on firm. The symbols ***, **, and * represent two-tailed p-values significant at the 0.01, 0.05 and 0.10 levels, respectively. Two-tailed t-statistics are presented in brackets. Untabulated control variables include ΔTAXAVOIDANCE, ΔETR_VOL, HAVEN, ΔMATERIALITY, NOL, ΔDTA_NET, ΔASSETS, ΔPTI, ΔLEV, BIG_4, ΔAUDITOR, BUSY, ΔNUM_KAMS, ΔINVREC, ΔMB, ΔSALES, LITIGATE, ΔFOREIGN, KAM_REGPERIOD, TAX_EXPERT, ΔFCF, MERGE_RESTRUCT, and PRIORLOSS

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**Conflict of interests** The authors have no relevant financial or nonfinancial interests to disclose.

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